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Sakamoto

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(54) **ZOOM LENS AND IMAGE PICKUP APPARATUS**

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(65) **Prior Publication Data**

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(30) **Foreign Application Priority Data**

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(57) **ABSTRACT**

(51) **Int. Cl.**

G02B 15/14 (2006.01)

G02B 15/167 (2006.01)

G02B 15/20 (2006.01)

A zoom lens includes, from an object side to an image side, a positive first lens unit configured not to move for zooming, at least three movable lens units configured to move for zooming, and a positive final lens unit closest to the image side in lens units and configured not to move for zooming. In the at least three movable lens units, a lens unit closest to the image side has a positive power, and a lens unit second closest to the image side has a negative power. The first lens unit includes a first sub unit closest to the object side and configured not to move for focusing and a second sub unit configured to move for focusing. The final lens unit includes a positive lens closest to the image side in elements having a refractive power in the final lens unit. Predetermined inequalities are satisfied.

(52) **U.S. Cl.**

CPC . **G02B 15/145125** (2019.08); **G02B 15/1461** (2019.08); **G02B 15/167** (2013.01); **G02B 15/20** (2013.01)

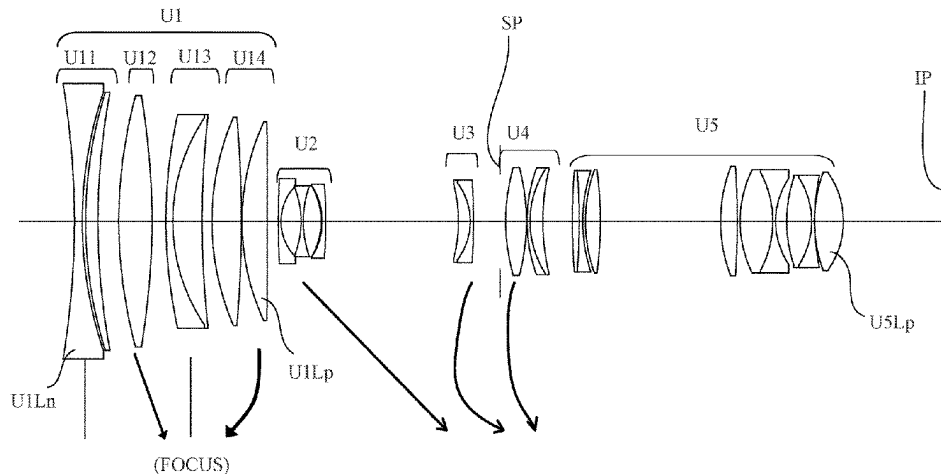
(58) **Field of Classification Search**

CPC G02B 15/145125; G02B 15/145127; G02B 15/1461; G02B 15/167; G02B 15/20

USPC 359/683, 684

See application file for complete search history.

18 Claims, 15 Drawing Sheets



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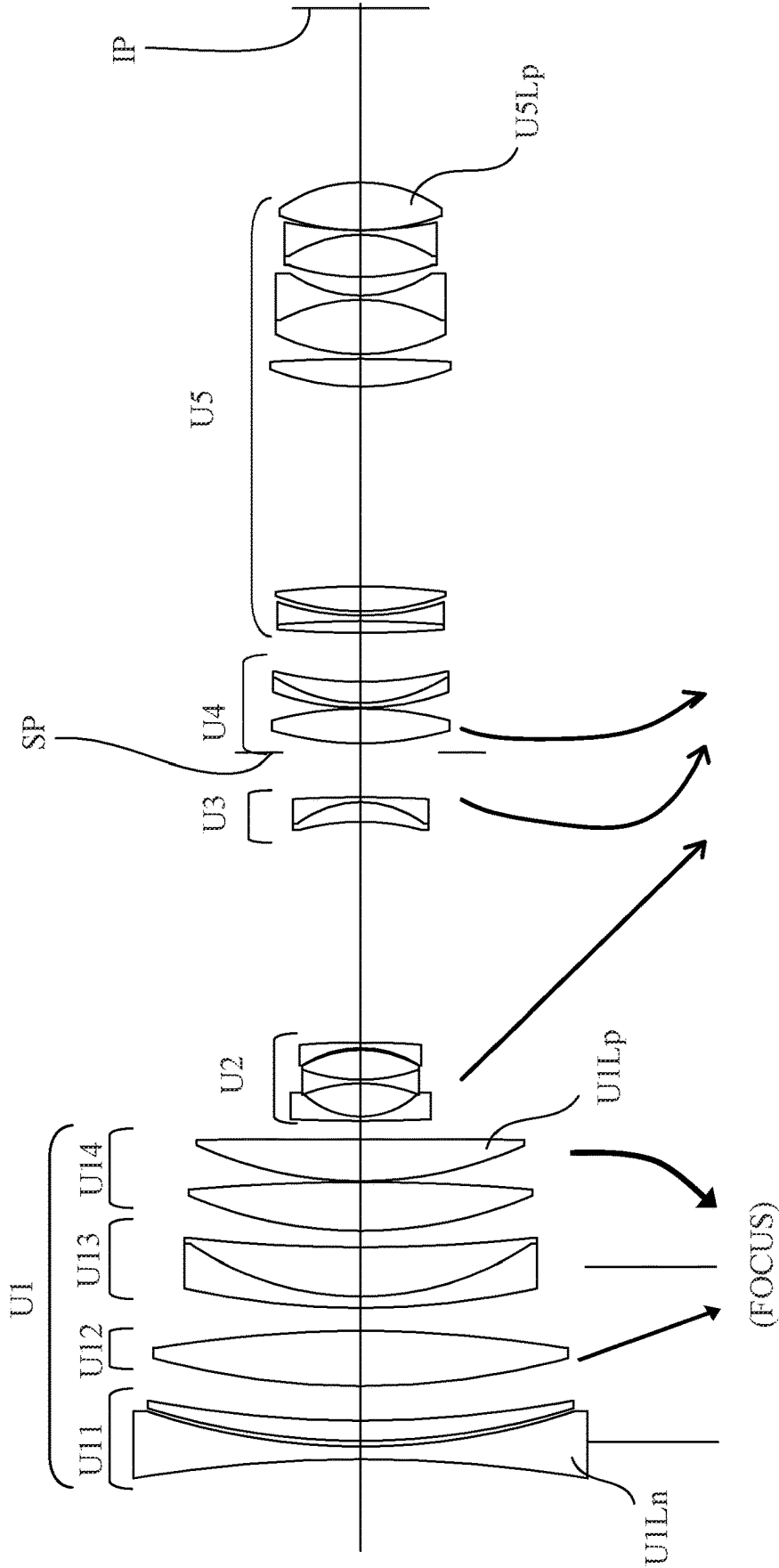


FIG. 1

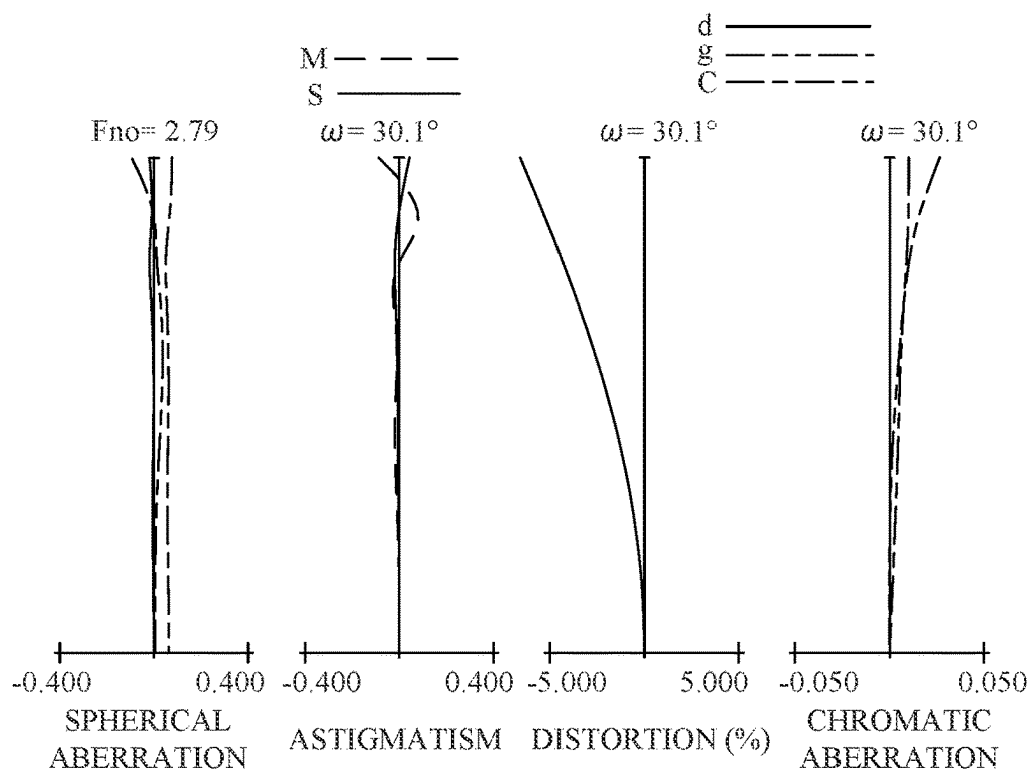


FIG. 2A

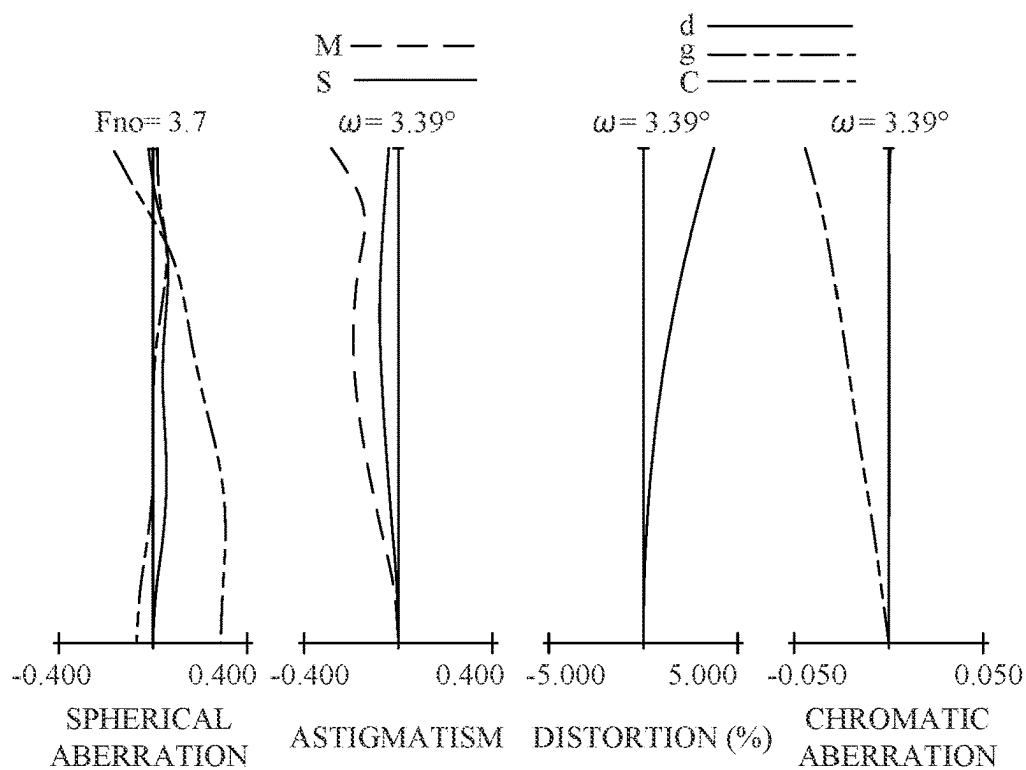


FIG. 2B

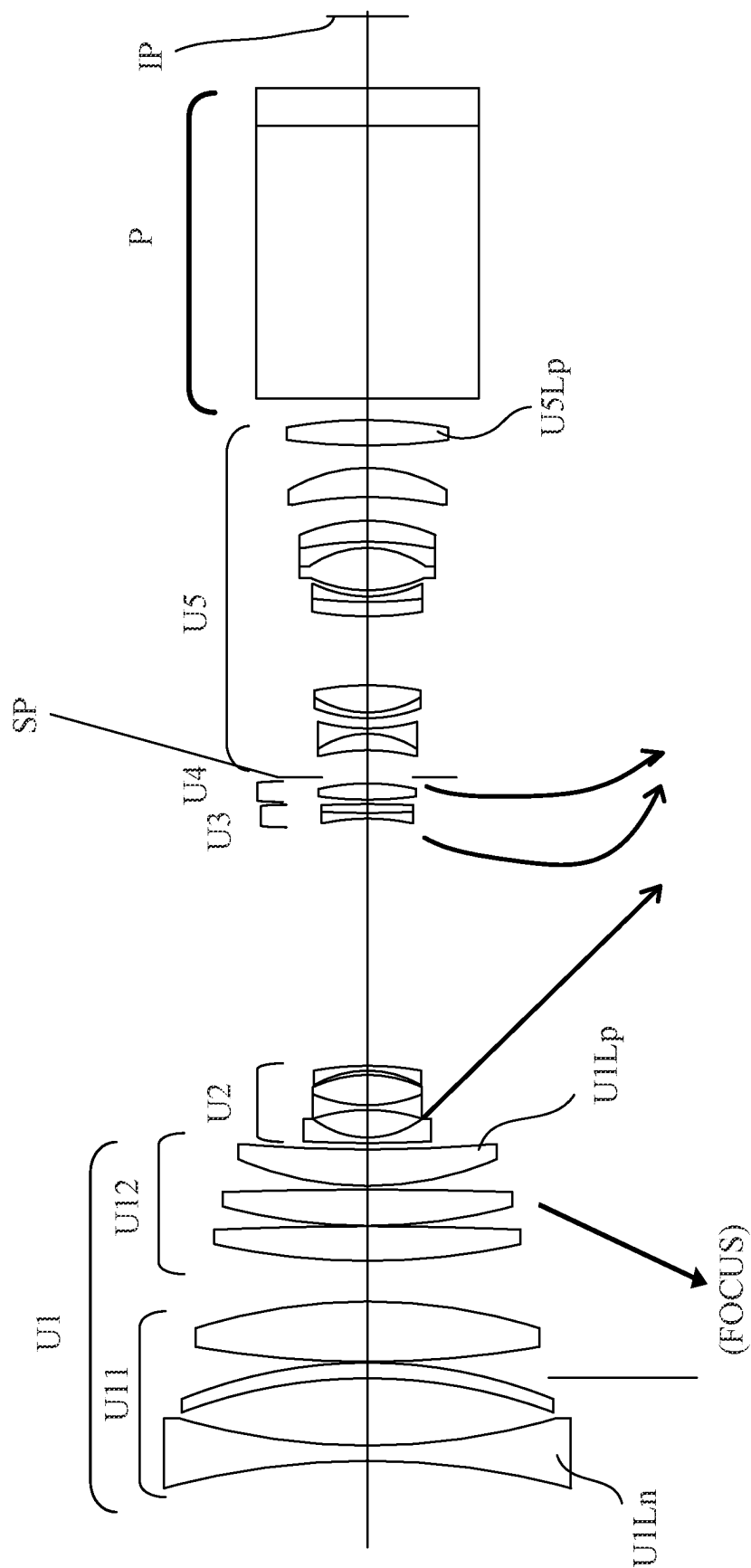


FIG. 3

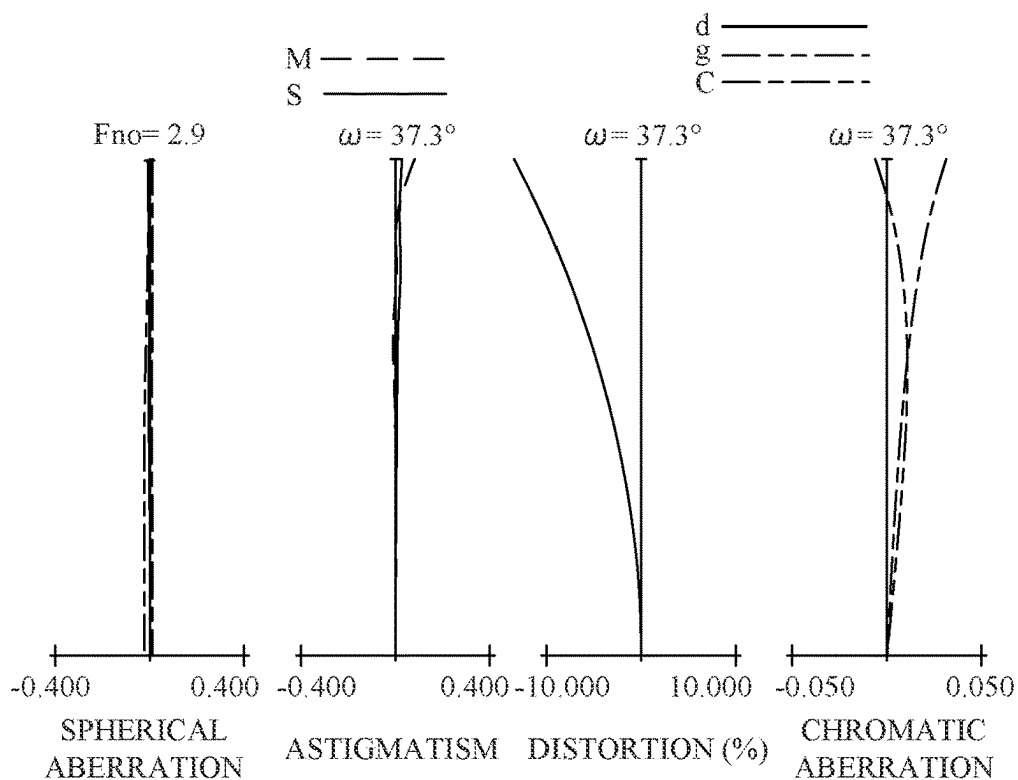


FIG. 4A

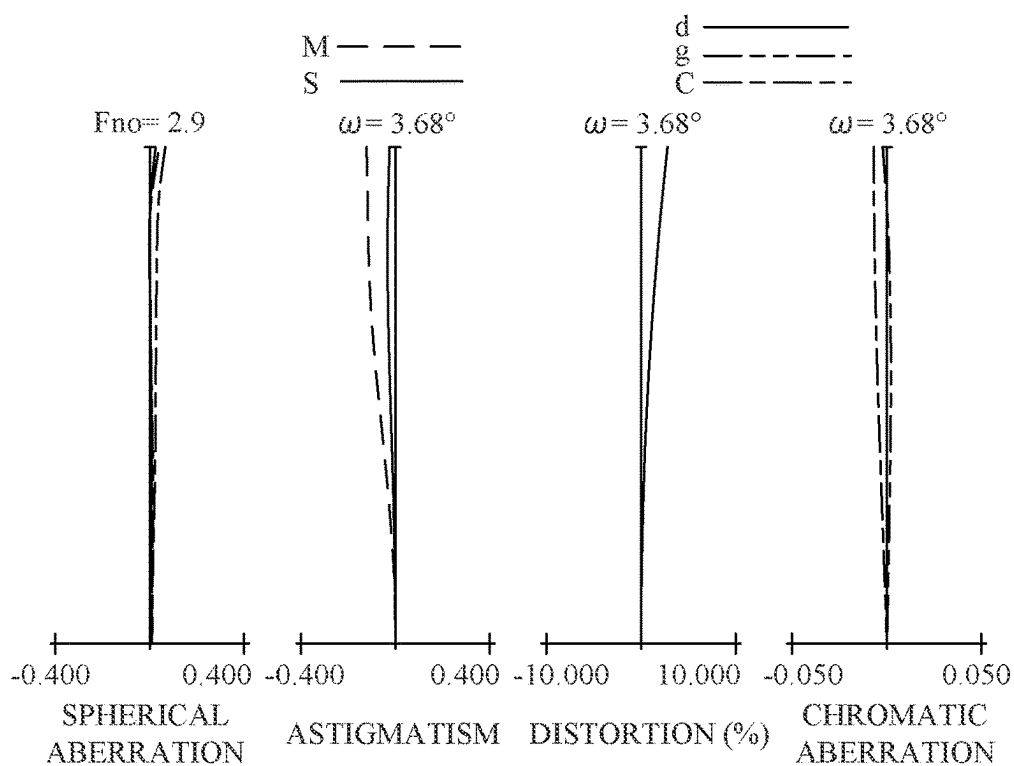


FIG. 4B

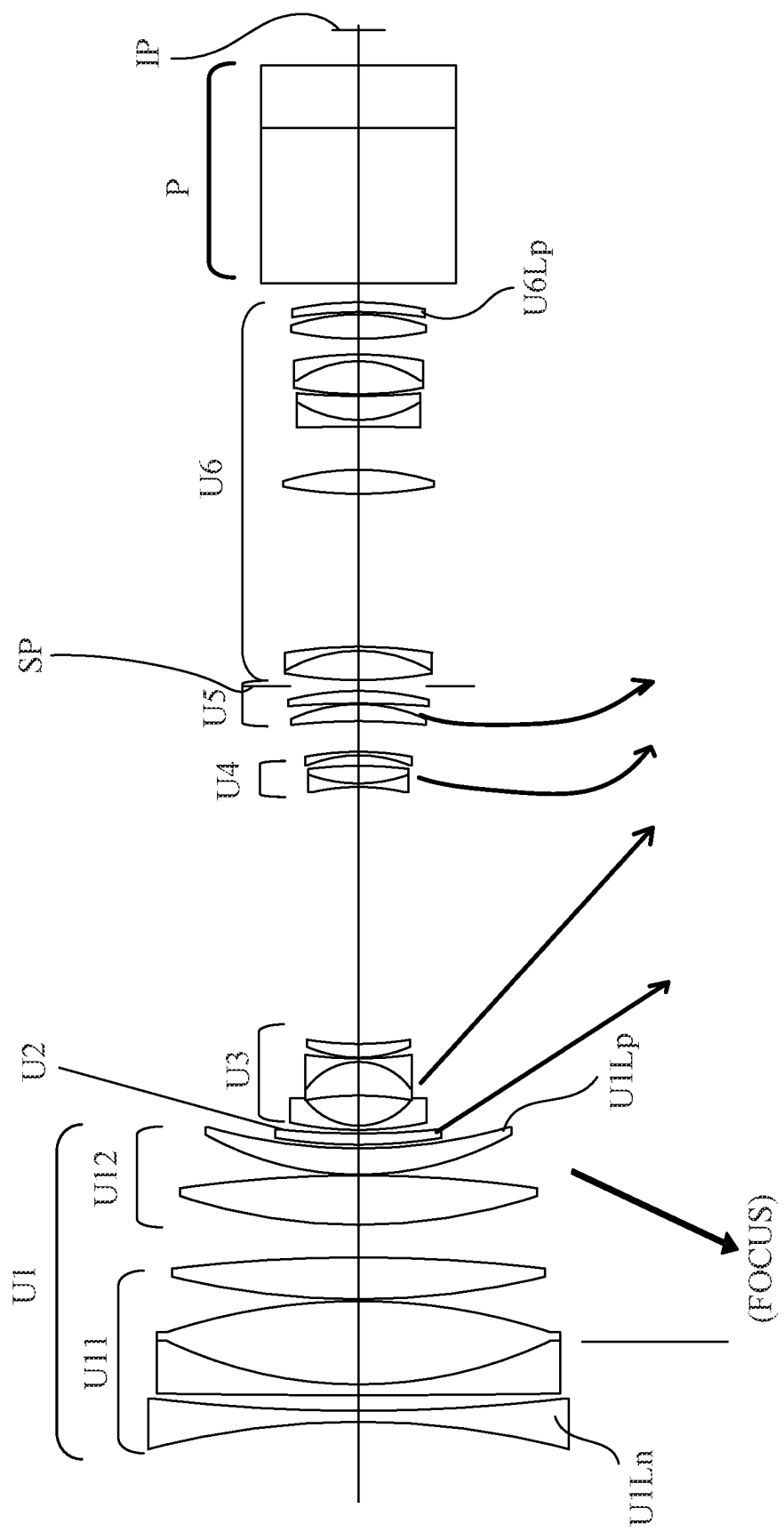


FIG. 5

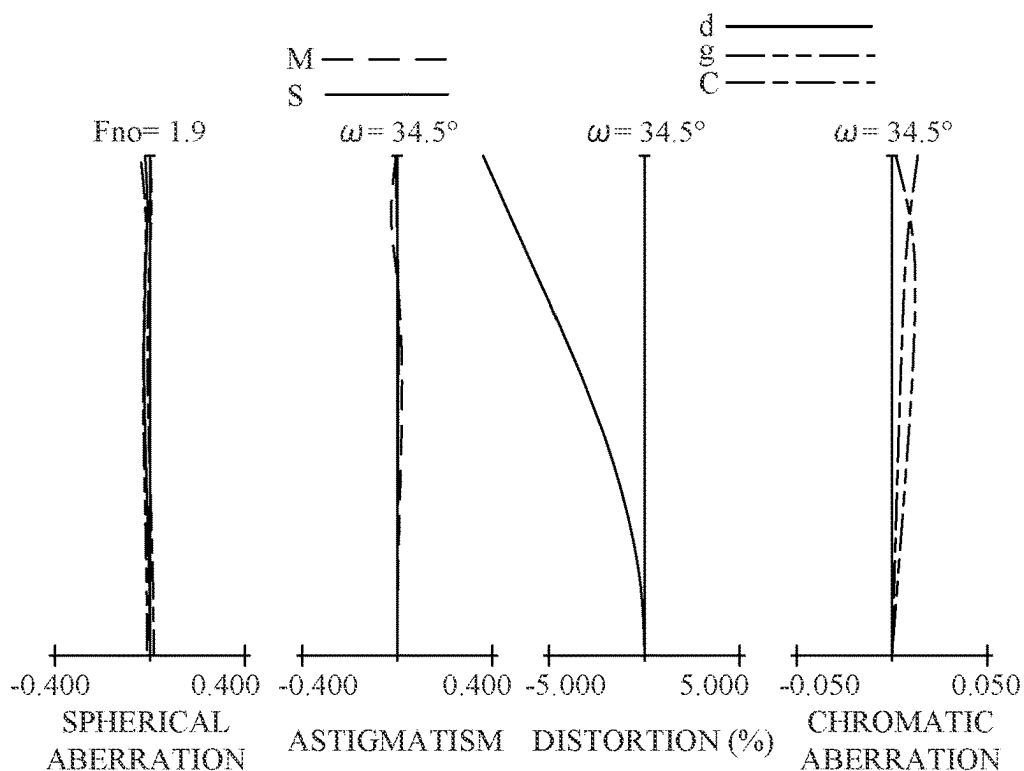


FIG. 6A

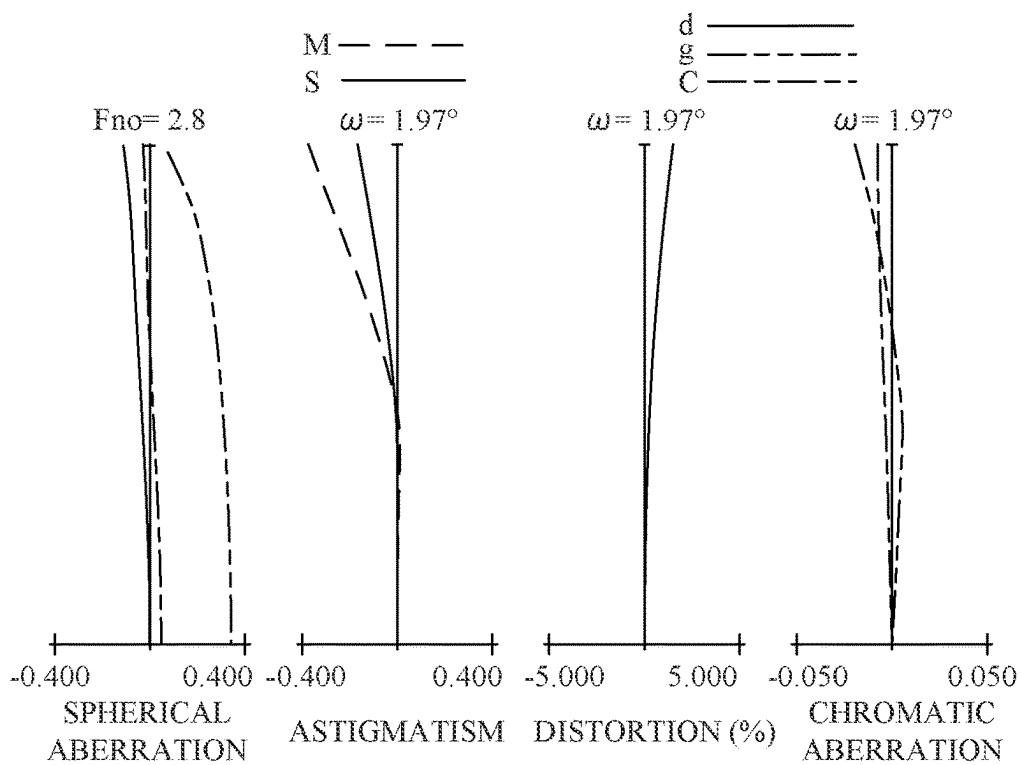
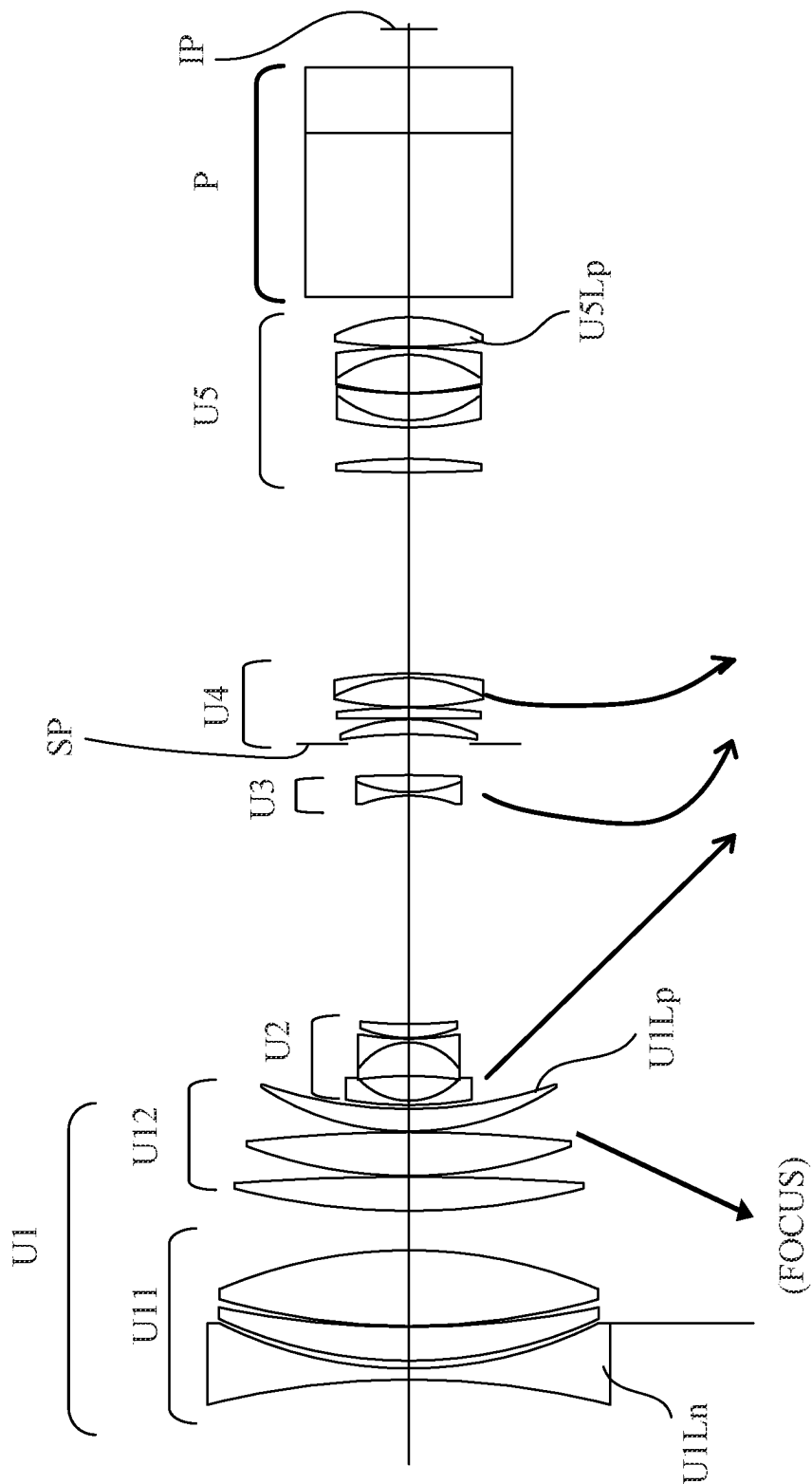


FIG. 6B



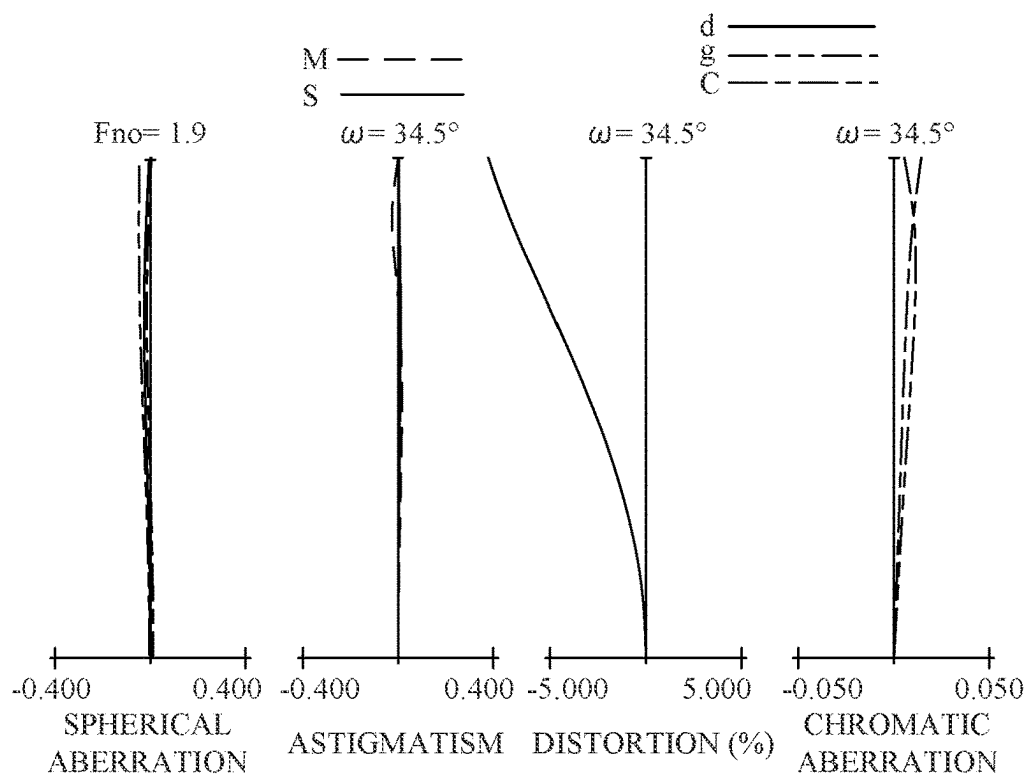


FIG. 8A

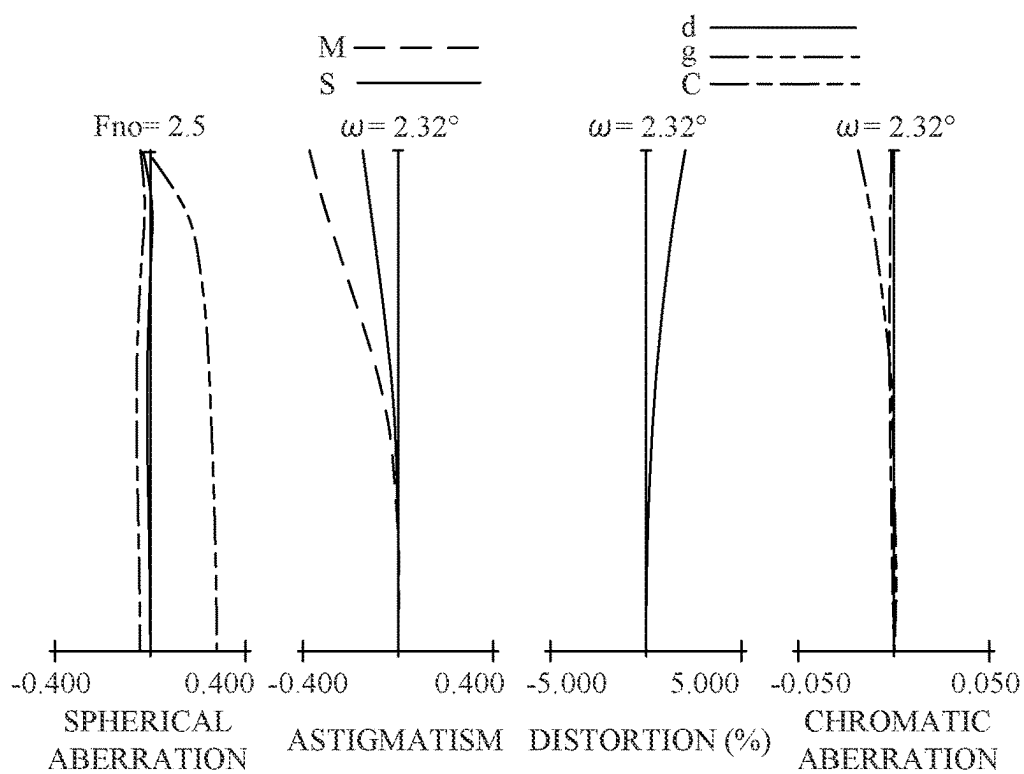


FIG. 8B

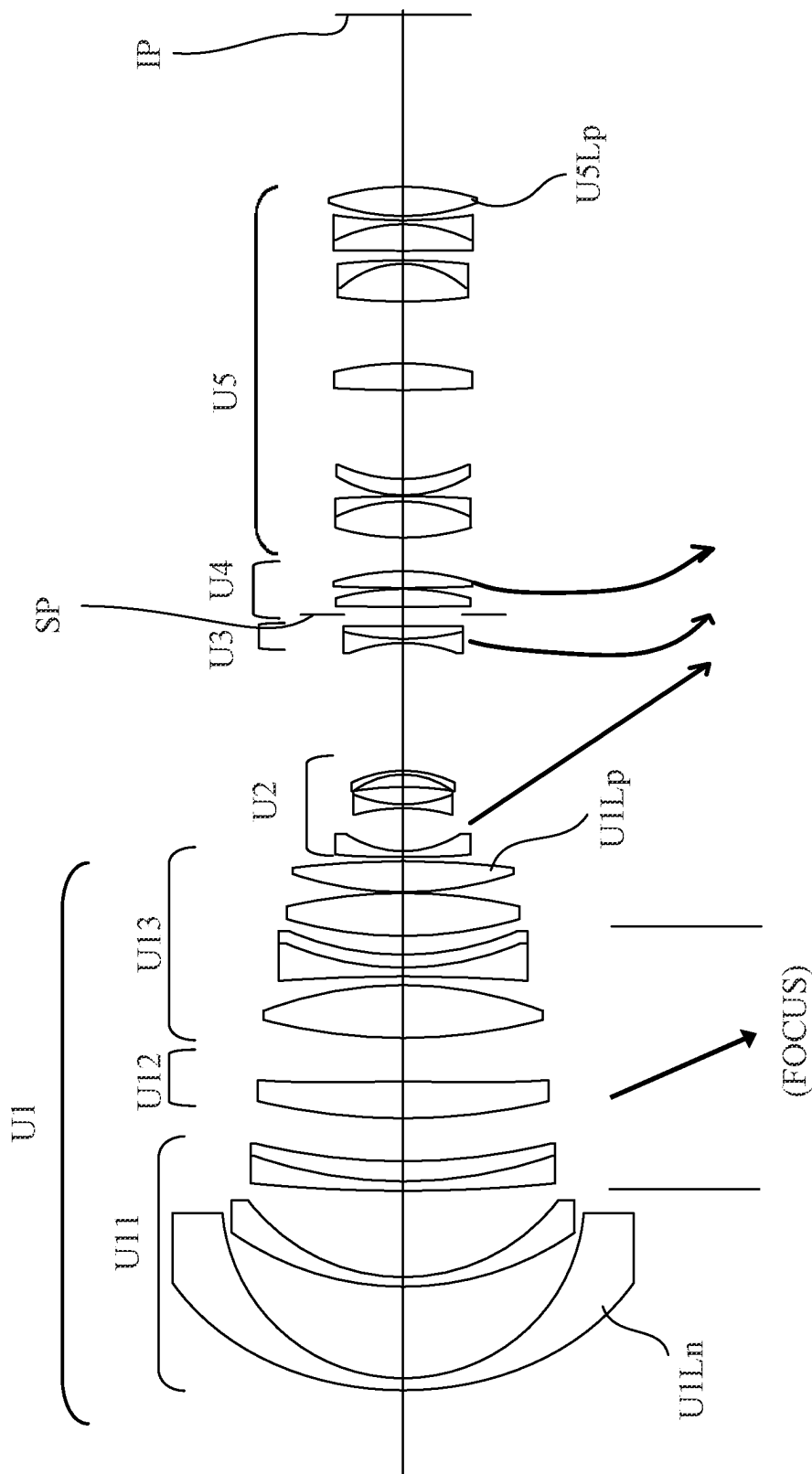


FIG. 9

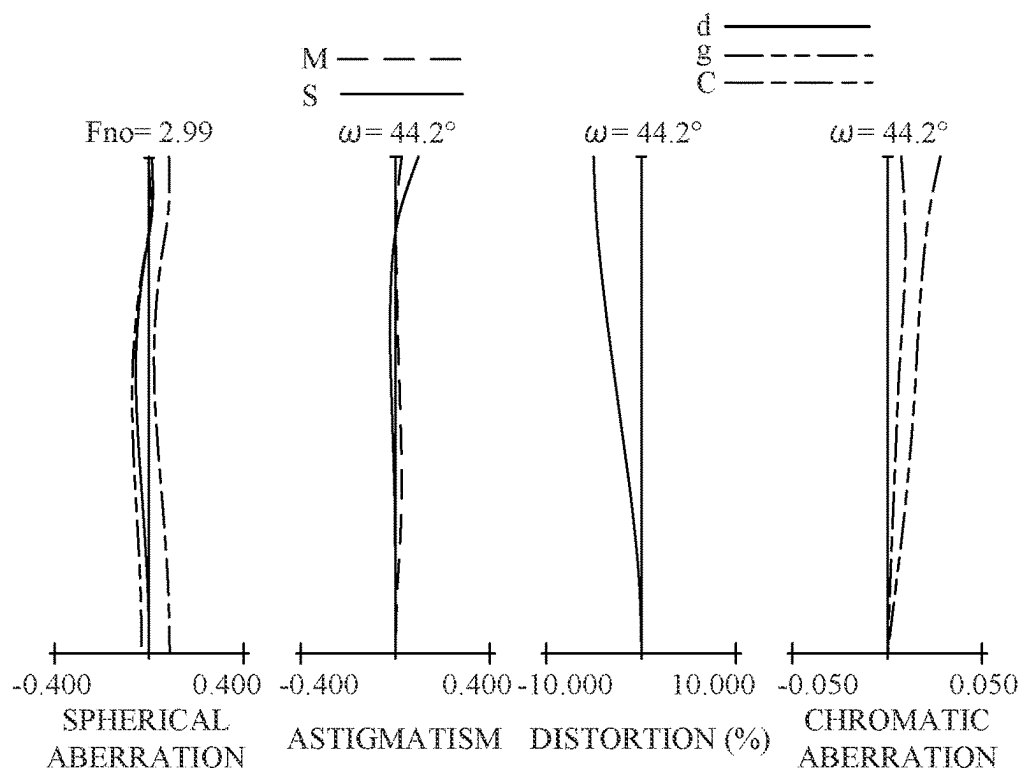


FIG. 10A

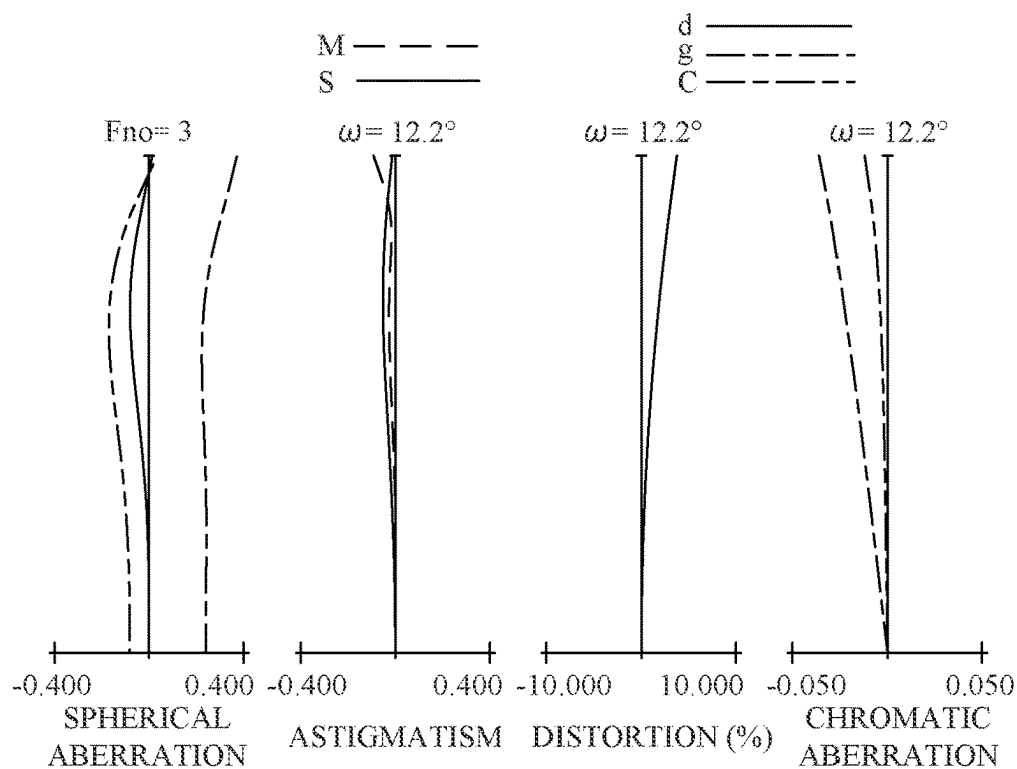


FIG. 10B

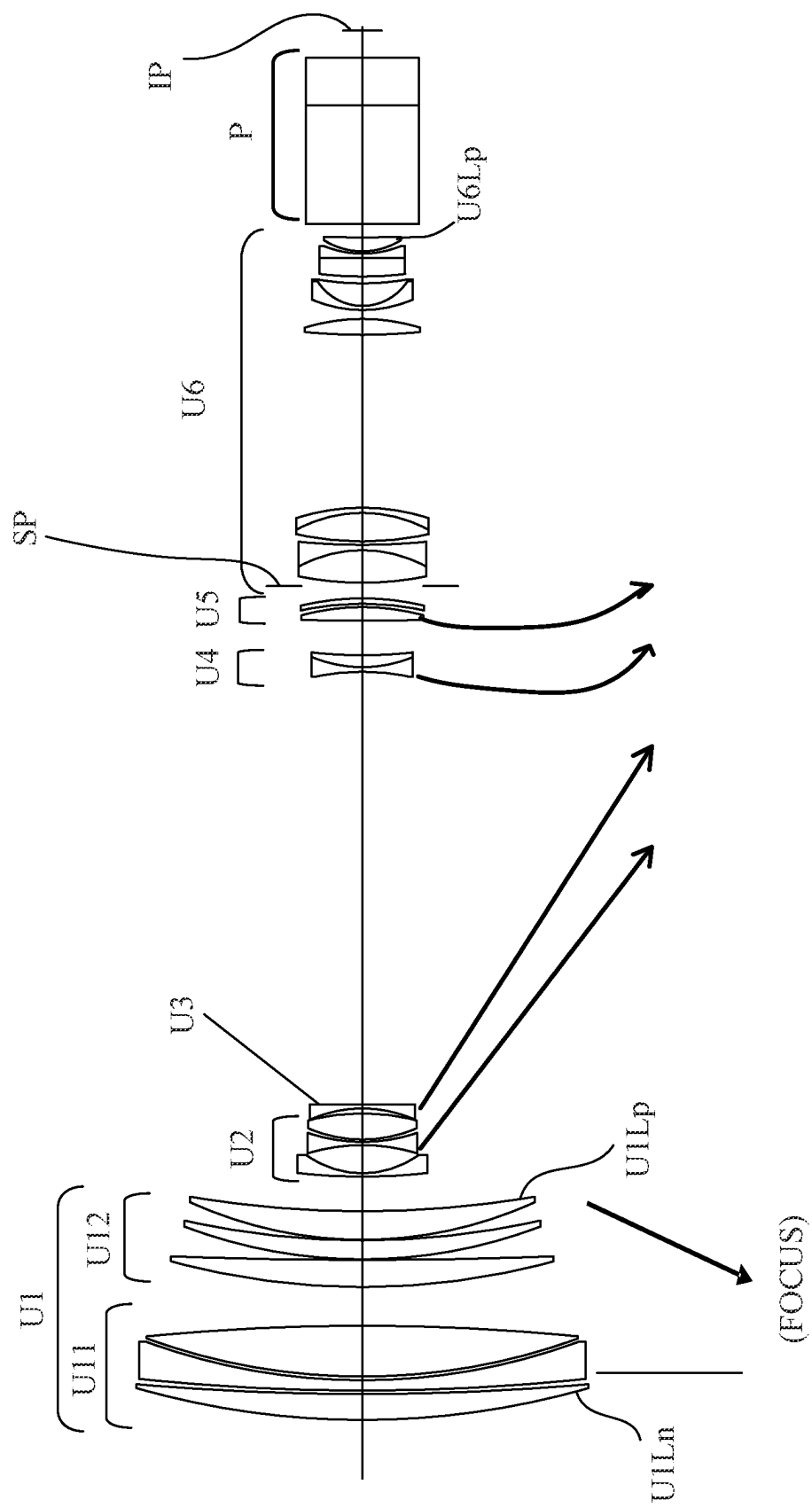


FIG. 11

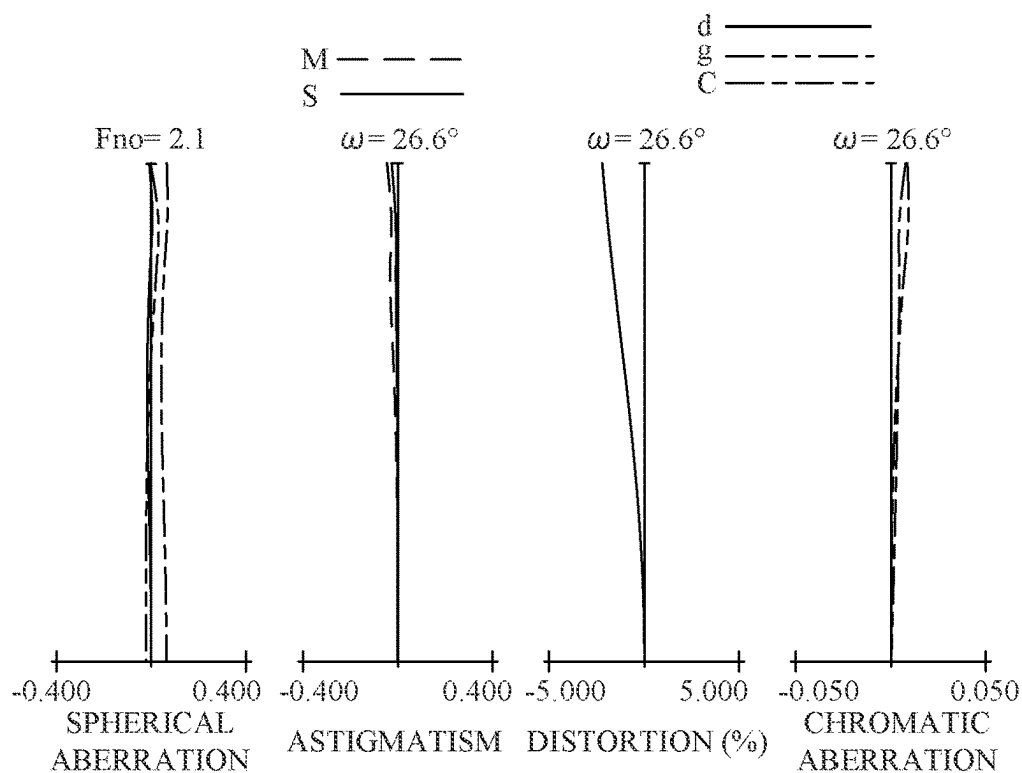


FIG. 12A

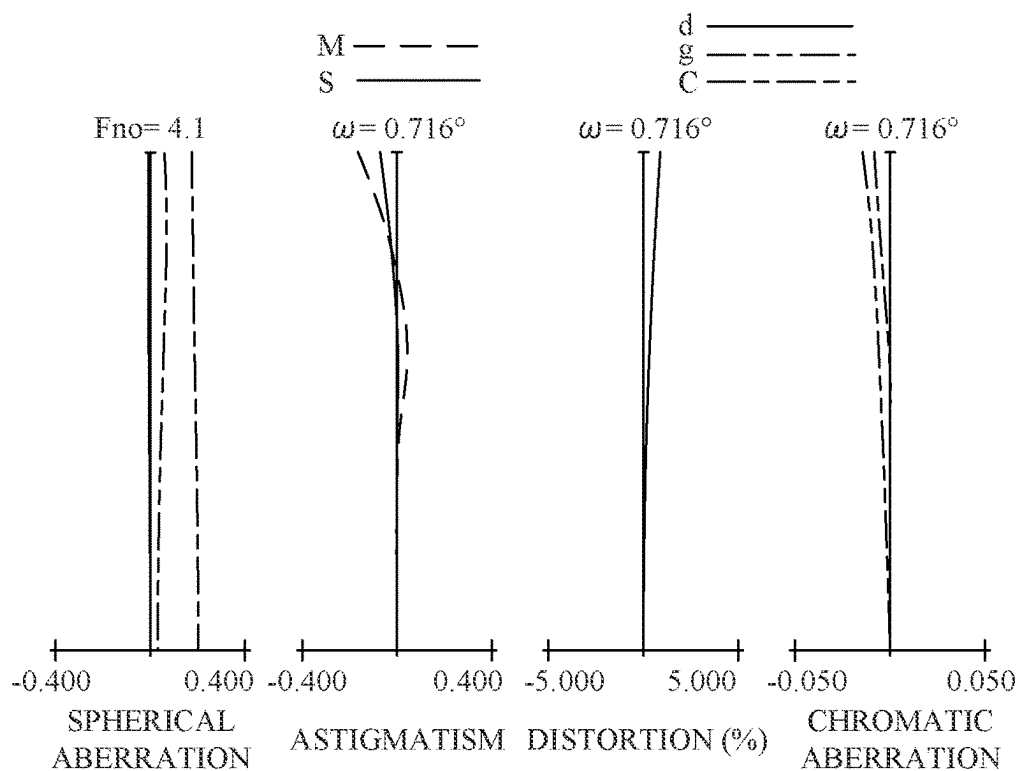


FIG. 12B

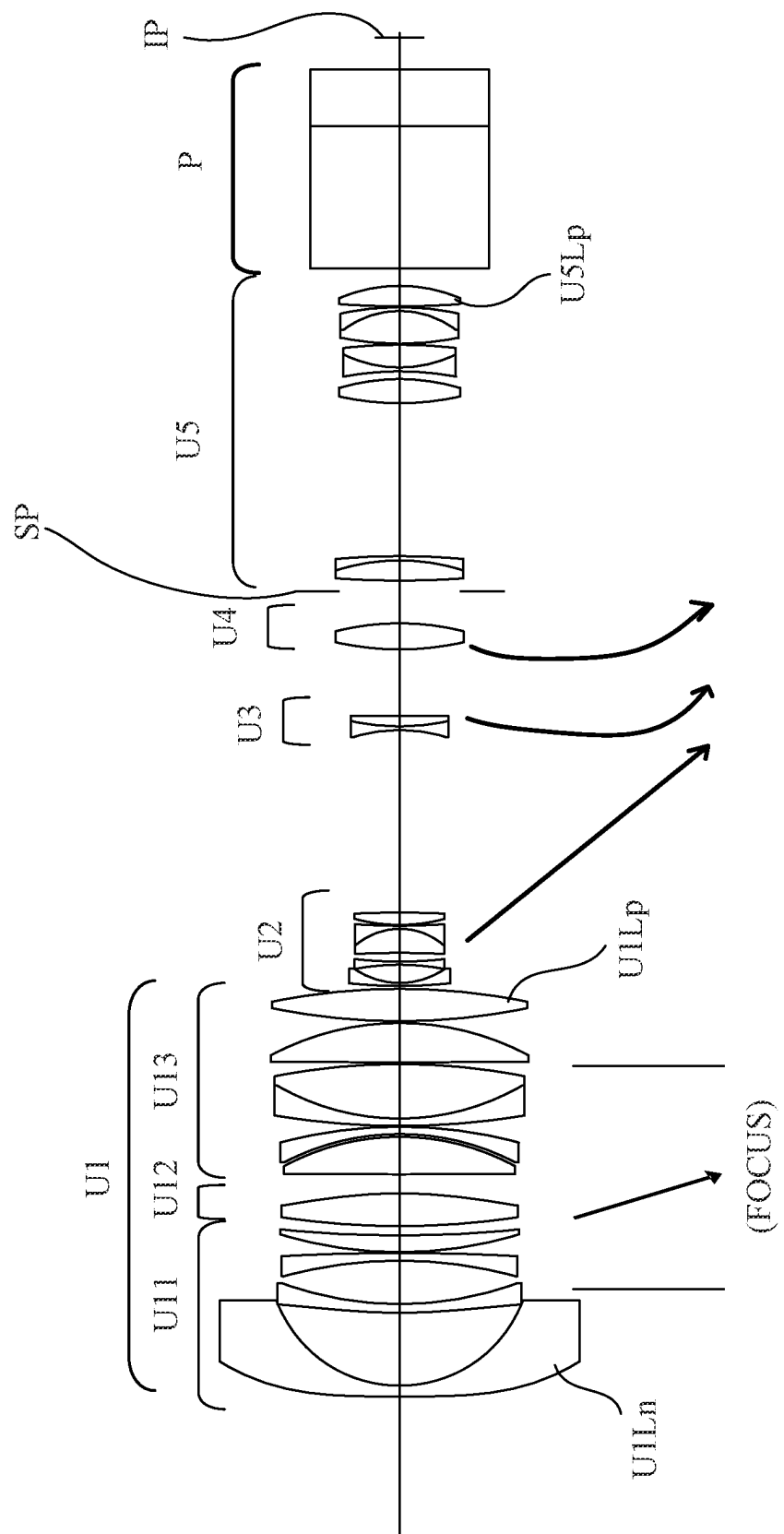


FIG. 13

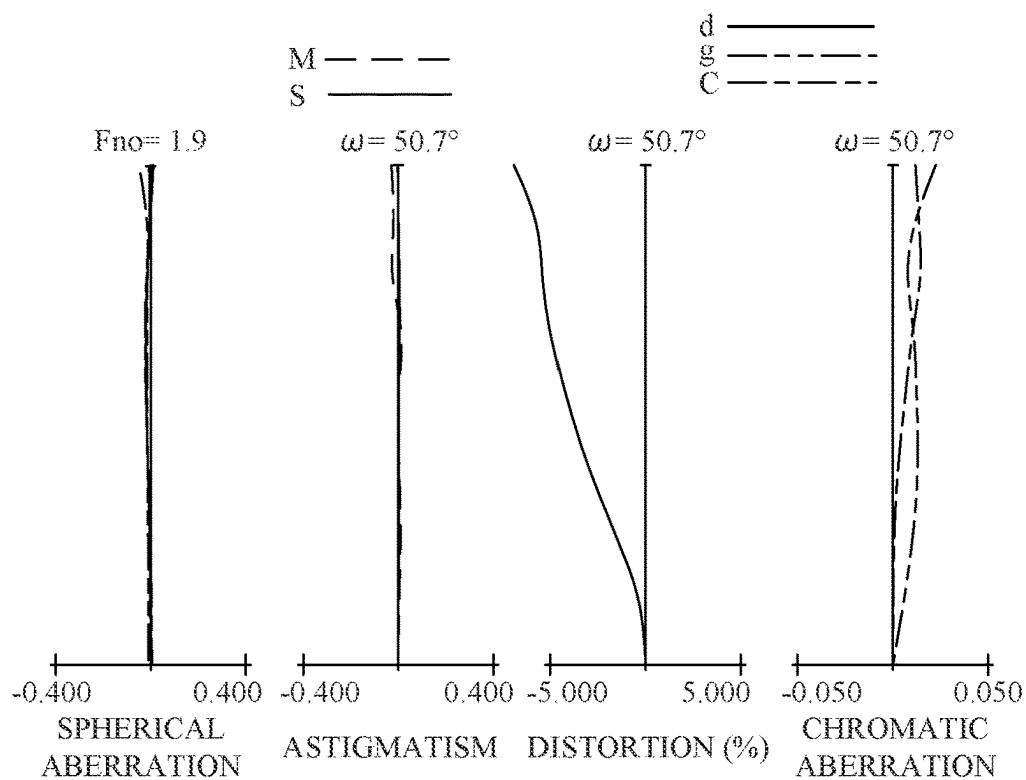


FIG. 14A

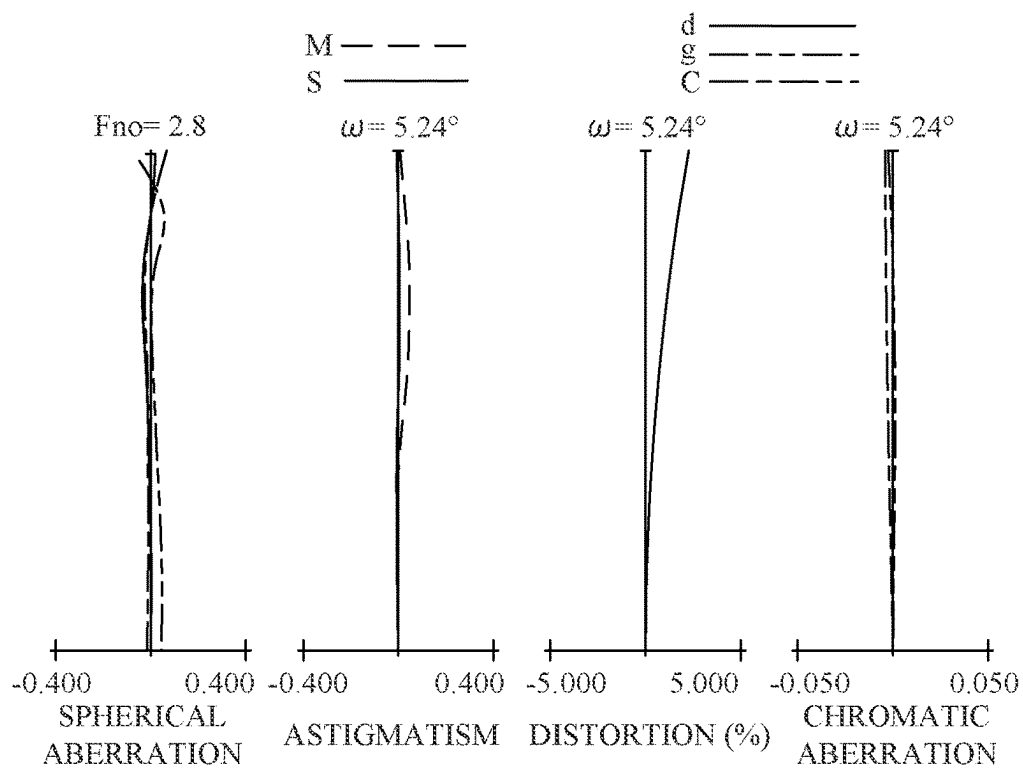


FIG. 14B

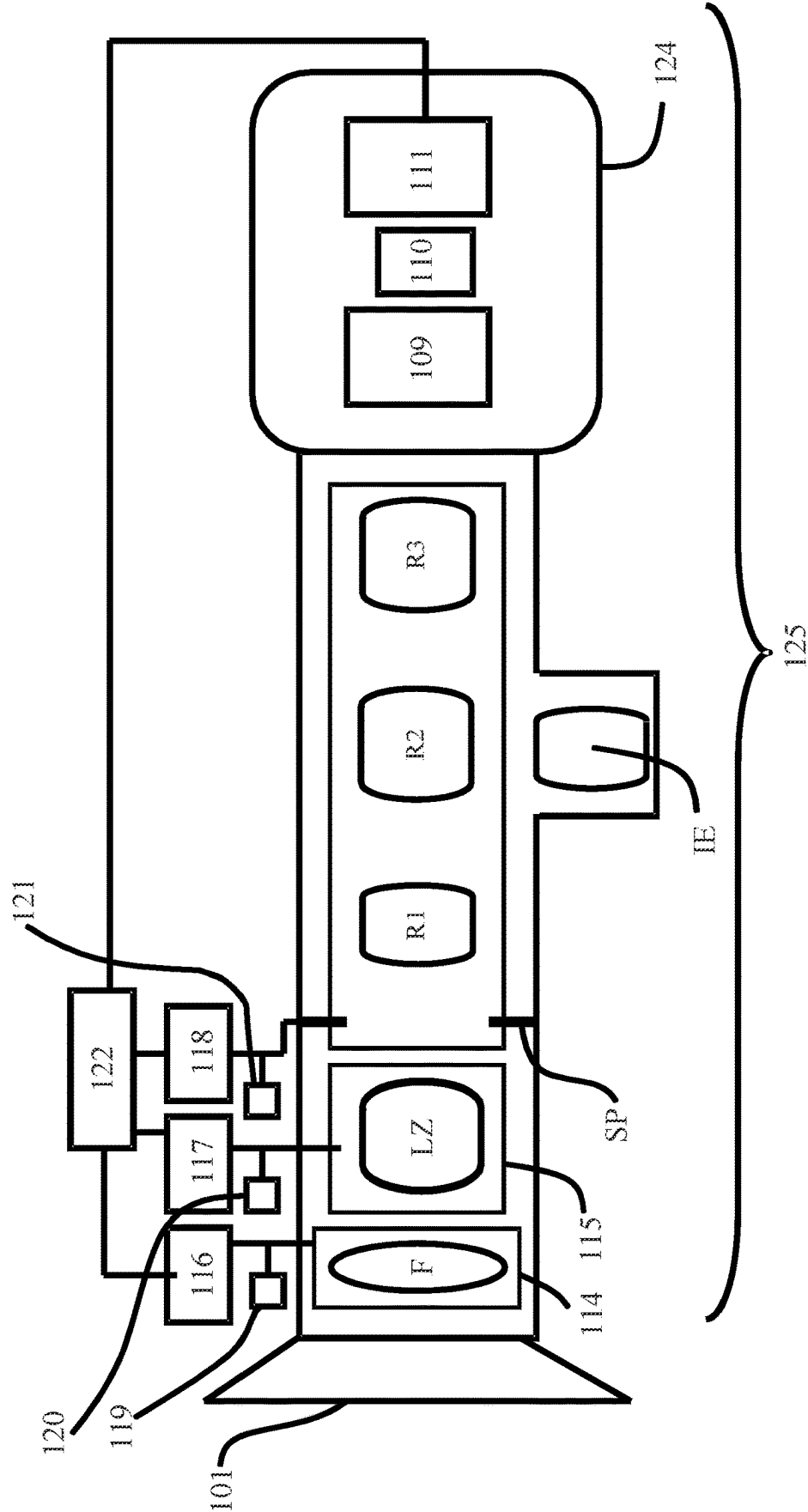


FIG. 15

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**ZOOM LENS AND IMAGE PICKUP
APPARATUS****BACKGROUND****Technical Field**

The aspect of the embodiments relates to a zoom lens and an image pickup apparatus.

Description of the Related Art

Zoom lenses are in demand to have high resolution uniformly from center to periphery of their angle of views and to cause less chromatic aberration. Japanese Patent Laid-Open No. 2016-012118 discloses a positive lead type zoom lens including, in order from an object side to an image side, a positive lens unit that does not move during zooming, a negative lens unit that moves during zooming, and a positive lens unit that moves during zooming.

Zoom lenses are also in demand to have wide angles, wide magnification ranges (high zoom ratios), high aperture ratios, and small sizes. Positive lead type zoom lenses are beneficial for increasing the zoom ratio as compared with negative lead type zoom lenses in each of which a first lens unit has a negative refractive index. However, in order that a positive lead type zoom lens is made to provide high optical performance, the positive lead type zoom lens is made to perform balanced correction on on-axis chromatic aberration and lateral chromatic aberration.

If the number of lenses is reduced so that a size of a zoom lens is reduced, and a refractive power of each lens unit or of a single lens is increased, chromatic aberration and other various errors may increase. If a wider angle of view and a higher zoom ratio are to be acquired, it may be difficult for the zoom lens to correct various aberrations, especially lateral chromatic aberration at a wide-angle end.

SUMMARY

A zoom lens according to one aspect of the embodiments includes, in order from an object side to an image side, a first lens unit having a positive refractive power and configured not to move for zooming, at least three movable lens units configured to move so that a distance between each pair of adjacent lens units therein changes for zooming, and a final lens unit disposed closest to the image side in lens units included in the zoom lens, having a positive refractive power, and configured not to move for zooming. A movable lens unit closest to the image side in the at least three movable lens units has a positive refractive power, and a movable lens unit second closest to the image side in the at least three movable lens units has a negative refractive power. The first lens unit includes a first sub unit closest to the object side and configured not to move for focusing, and a second sub unit configured to move for focusing. The final lens unit includes a positive lens closest to the image side in elements having a refractive power in the final lens unit. Following inequalities are satisfied:

$$62 \leq vdp \leq 81.54,$$

$$2.24 \leq Ndp + 0.01 vdp, \text{ and}$$

$$4 \leq Z \leq 50$$

where vdp represents an Abbe number with respect to a d-line of the positive lens, Ndp represents a refractive index

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with respect to the d-line of the positive lens, and Z represents a zoom ratio of the zoom lens.

An apparatus including the above zoom lens also constitute another aspect of the embodiments of the present disclosure.

Further features of the disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a sectional view of a zoom lens according to Example 1 (Numerical Example 1).

FIGS. 2A and 2B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 1.

FIG. 3 is a sectional view of a zoom lens according to Example 2 (Numerical Example 2).

FIGS. 4A and 4B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 2.

FIG. 5 is a sectional view of a zoom lens according to Example 3 (Numerical Example 3).

FIGS. 6A and 6B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 3.

FIG. 7 is a sectional view of a zoom lens according to Example 4 (Numerical Example 4).

FIGS. 8A and 8B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 4.

FIG. 9 is a sectional view of a zoom lens according to Example 5 (Numerical Example 5).

FIGS. 10A and 10B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 5.

FIG. 11 is a sectional view of a zoom lens according to Example 6 (Numerical Example 6).

FIGS. 12A and 12B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 6.

FIG. 13 is a sectional view of a zoom lens according to Example 7 (Numerical Example 7).

FIGS. 14A and 14B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 7.

FIG. 15 is a diagram illustrating an image pickup apparatus having the zoom lens according to any one of Examples 1 to 7.

DESCRIPTION OF THE EMBODIMENTS

Referring now to the accompanying drawings, a description will be given of embodiments according to the present disclosure.

First, prior to a specific description of Examples 1 to 7, a description is given of matters common to each example. A zoom lens according to each example is suitable as an image pickup optical system for an image pickup apparatus such as a still camera, a video camera, a TV camera, a cinema camera, and a monitoring camera. In the zoom lens according to each example, a lens unit or a sub unit is a group of one or a plurality of lenses that integrally move as a whole during zooming between a wide-angle end and a telephoto end and during focusing to an infinite distance side or to a close side. That is, a distance between each pair of adjacent lens units or sub units changes during zooming and focusing.

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A lens unit may include an aperture diaphragm. The wide-angle end represents a zooming state with a maximum angle of view (shortest focal length), and the telephoto end represents a zooming state with a minimum angle of view (maximum focal length). Each lens unit that moves during zooming is located, in the state of the wide-angle end, at one end and, in the state of the telephoto end, at the other end of a mechanical or control movable range on an optical axis.

Next, a description is given of the zoom lens illustrated in FIG. 1 according to Example 1 as an example (with reference signs in parentheses).

The zoom lens according to each example includes, in order from an object side to an image side, a first lens unit (U1) that has a positive refractive power and does not move (is fixed) during (or for) zooming, at least three movable lens units (U2 to U4) that move so that a distance between each pair of adjacent lens units therein changes during (or for) zooming, and a final lens unit (U5) that is closest to the image side, has a positive refractive power, and does not move during (or for) zooming. In the above-described at least three movable lens units (U2 to U4), a movable lens unit (U4) closest to the image side has a positive refractive power, and a movable lens unit (U3) second closest to the image side has a negative refractive power. The first lens unit (U1) includes a fixed sub unit (U11) that is closest to the object side and does not move during (or for) focusing, and a movable sub unit (U12 and U14) that moves during (or for) focusing.

In this way, the zoom lens according to each example performs zooming by moving the at least three movable lens units (U2 to U4), increases a zoom ratio by using the negative movable lens unit (U3) second closest to the image side in the at least three movable lens units (U2 to U4), and effectively reduces (or corrects) an aberration variation during zooming by using the positive movable lens unit (U4) closest to the image side in the at least three movable lens units (U2 to U4). This makes it possible to realize a small-sized zoom lens that has a wide magnification range (high zoom ratio) and high optical performance.

In addition, by moving a part of the first lens unit (U1) during focusing and disposing the fixed sub unit (U11) at a position closest to the object side in the first lens unit (U1), it is possible to reduce an aberration variation caused by focusing. Further, it is possible to reduce sizes of the movable sub unit (U12, U14) that moves during focusing and of its driving mechanism, and thereby both high zoom ratio and size reduction can be realized.

The zoom lens according to each example further satisfies the following inequalities (1) to (3) in addition to the above-described configurations of the movable lens units and the first lens unit, so as to realize a zoom lens suitable for improving a definition and a function of an image pickup apparatus.

In the final lens unit (U5), a lens closest to the image side in elements having refractive powers is a positive lens (U5Lp). vdp and Ndp respectively represent an Abbe number with respect to a d-line and a refractive index with respect to the d-line of the positive lens U5Lp, and Z represents the zoom ratio of the zoom lens. Here, the zoom lens according to each example satisfies the following inequalities (1) to (3).

$$62 \leq v_d \quad (1)$$

$$2.24 \leq N_{dp} + 0.01 v_{dp} \quad (2)$$

$$4 \leq Z \leq 5 \quad (3)$$

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The inequalities (1) and (2) represents conditions relating to characteristics of optical glass of the positive lens (U5Lp) closest to the image side in the final lens unit U5. Usually, optical glass contains a plurality of types of metal oxides such as SiO₂, TiO₂, La₂O₃, Al₂O₃, Nb₂O₅, ZrO₂, and Gd₂O₃. Among them, TiO₂ has an effect of increasing a refractive index and reducing an Abbe number. Therefore, optical glass with a high content of TiO₂ has a relatively high refractive index and a high dispersion. Gd₂O₃ has an effect of increasing a refractive index and increasing an Abbe number. Optical glass with a high content of Gd₂O₃ has a relatively high refractive index and a low dispersion. TiO₂ originally has a high refractive index and a high dispersion and Gd₂O₃ originally has a high refractive index and a low dispersion, and therefore the characteristics of optical glass containing them are similar to the original characteristics of the metal oxides.

As described above, optical glass has a property such that its characteristics change depending on contents of substances (components), and thus desired optical characteristics (refractive index and Abbe number) can be acquired by properly setting amounts of the substances. The same similarly applies to optical ceramics. For example, if optical ceramics includes a high content of a substance having a high refractive index and a low dispersion, the optical ceramics have relatively high refractive index and low dispersion. Therefore, it is possible to acquire an optical material such as optical glass and optical ceramics having desired optical characteristics if amounts of substances and metal oxides such as SiO₂, TiO₂, and La₂O₃ are properly set and are dissolved or sintered.

In the zoom lens having the above configurations, as the angle of view becomes wider, a refractive power of a negative lens (U1Ln) closest to the object side in the first lens unit U1 becomes stronger. Further, at the wide-angle end, a height increases of an off-axis principal ray entering the negative lens (U1Ln) closest to the object side in the first lens unit U1, and as a result, a generated amount of lateral chromatic aberration increases. Thus, the wider angle of view makes it difficult to correct the lateral chromatic aberration especially at the wide-angle end.

Further, as the zoom ratio becomes higher, a height of an on-axis ray in the first lens unit (U1) increases in proportion to a focal length, and as a result, on-axis chromatic aberration occurring in the first lens unit (U1) increases. In order that the on-axis chromatic aberration occurring in the first lens unit (U1) is reduced, optical glass having a high anomalous dispersion in a movable lens unit that has a strong negative refractive power and moves during zooming is used. A height of an off-axis principal ray entering the movable lens unit at the wide-angle end is high, and therefore large lateral chromatic aberration is caused by the optical glass having the high anomalous dispersion in the movable lens unit. Thus, increasing the zoom ratio also makes it difficult to correct the lateral chromatic aberration at the wide-angle end.

In general, a lens placed at a position at which a height of an off-axis ray is high has a large contribution to correction of lateral chromatic aberration. For correcting well the lateral chromatic aberration at the wide-angle end that occurs with widening the angle of view and increasing the zoom ratio, it is effective to use an optical material having a low dispersion and a high anomalous dispersion in the positive lens (U5Lp) at a position farthest from the aperture. In this positive lens (U5Lp), a chief paraxial ray, which has a large effect on lateral chromatic aberration, passes through a position where a height from the optical axis is high.

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Therefore, using the above-described optical material for the positive lens (U5Lp) makes larger the effect on lateral chromatic aberration, and thus the lateral chromatic aberration can be corrected well.

The inequality (1) represents a condition relating to the Abbe number of the positive lens (U5Lp) closest to the image side in the final lens unit (U5). If v_{dp} is smaller than the lower limit of the inequality (1), it is difficult to correct lateral chromatic aberration that occurs in the first lens unit (U1) and is insufficiently corrected by the first lens unit (U1). In addition, such an optical glass material makes the positive lens (U5Lp) have a small partial dispersion ratio $\theta_g F$, which makes it difficult to correct a secondary spectrum of lateral chromatic aberration occurring in a movable lens unit having a strong negative refractive power that moves during zooming.

The numerical range of the inequality (1) may be set to a numerical range of the following inequality (4).

$$80 \leq v_{dp} \quad (4)$$

The numerical range of the inequality (4) may be set to the following numerical range.

$$80 \leq v_{dp} \leq 97 \quad (4a)$$

If v_{dp} is larger than the upper limit of the inequality (4a), it is difficult to produce a low-dispersion optical glass material.

The inequality (2) represents a condition relating to a relationship between the Abbe number and the refractive index of the positive lens (U5Lp). If $N_{dp} + 0.01 v_{dp}$ is smaller than the lower limit of the inequality (2), such an optical glass material makes the positive lens (U5Lp) have a small partial dispersion ratio $\theta_g F$, which makes it difficult to correct a secondary spectrum of lateral chromatic aberration occurring in a movable lens unit having a strong negative refractive power that moves during zooming.

The numerical range of the inequality (2) may be set to the following numerical range.

$$2.24 \leq N_{dp} + 0.01 v_{dp} \leq 2.50 \quad (2a)$$

If $N_{dp} + 0.01 v_{dp}$ is larger than the upper limit of the inequality (2a), it is difficult to produce an optical glass material having a low dispersion and a high refractive index.

The numerical range of the inequality (2) may be set to the following numerical range.

$$2.24 \leq N_{dp} + 0.01 v_{dp} \leq 2.40 \quad (2b)$$

The inequality (3) represents a condition for realizing a zoom lens having a large zoom ratio while having a small size. If Z is smaller than the lower limit of the inequality (3), it is difficult to increase the zoom ratio of the zoom lens. If Z is larger than the upper limit of the inequality (3), it is difficult to reduce the size of the zoom lens.

The numerical range of the inequality (3) may be set to the following numerical range.

$$4.3 \leq Z \leq 47.0 \quad (3a)$$

The numerical range of the inequality (3) may be set to the following numerical range.

$$4.4 \leq Z \leq 43.0 \quad (3b)$$

If the above-described conditions (inequalities) are satisfied, it is possible to realize a small-sized and light-weighted zoom lens having a wide angle of view, a large diameter, a high zoom ratio, and good optical performance.

The zoom lens according to each example may satisfy at least one of the following conditions (inequalities). In the zoom lens according to each example, the final lens unit

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(U5) may include four or more lenses each of which has a refractive power. When the number of lenses included in the final lens unit (U5) is properly set, it is possible to correct well various aberrations at the wide-angle end. The final lens unit (U5) may include at least six or more lenses each of which has a refractive power.

The zoom lens according to each example may satisfy the following inequality (5) where f_p represents a focal length of the positive lens (U5Lp) closest to the image side in the final lens unit (U5), and f_v represents a focal length of the lens unit (U2) having the strongest negative refractive power in the at least three movable lens units (U2 to U4).

$$0.5 \leq |f_p/f_v| \leq 50.0 \quad (5)$$

If $|f_p/f_v|$ is larger than the upper limit of the inequality (5), the refractive power of the positive lens (U5Lp) is so weak that it becomes difficult to correct the lateral chromatic aberration at the wide-angle end. If $|f_p/f_v|$ is smaller than the lower limit of the inequality (5), the refractive power of the positive lens (U5Lp) is so strong that the lateral chromatic aberration at the wide-angle end is excessively corrected. Further, since the refractive power of the positive lens (U5Lp) is strong, various aberrations occurring in the final lens unit (U5) increase, which makes it difficult to correct various aberrations in the wide-angle end.

The numerical range of the inequality (5) may be set to the following numerical range.

$$0.7 \leq |f_p/f_v| \leq 40.0 \quad (5a)$$

The numerical range of the inequality (5) may be set to the following numerical range.

$$0.85 \leq |f_p/f_v| \leq 35.00 \quad (5b)$$

The zoom lens according to each example may satisfy the following inequality (6) where L_e represents a distance on the optical axis from a lens surface closest to the object side in the final lens unit (U5) to a lens surface closest to the image side in the final lens unit (U5), and f_w represents a focal length at the wide-angle end of the zoom lens.

$$2.5 \leq L_e/f_w \leq 25.0 \quad (6)$$

If L_e/f_w is larger than the upper limit of the inequality (6), a height of an off-axis ray at the positive lens (U5Lp) is so high that it becomes difficult to reduce the size of the positive lens (U5Lp). If L_e/f_w is smaller than the lower limit of the inequality (6), the height of the off-axis ray at the positive lens (U5Lp) is so low that it becomes difficult to correct the lateral chromatic aberration at the wide-angle end.

The numerical range of the inequality (6) may be set to the following numerical range.

$$3.0 \leq L_e/f_w \leq 20.0 \quad (6a)$$

The numerical range of the inequality (6) may be set to the following numerical range.

$$3.5 \leq L_e/f_w \leq 17.0 \quad (6b)$$

In the zoom lens according to each example, the movable lens unit (U2) having the strongest negative refractive power in the at least three movable lens units (U2 to U4) each of which moves during zooming may include at least two negative lenses and one positive lens. Thereby, it is possible to realize a small zoom lens that has a high zoom ratio and high optical performance and is suitable for improving a definition and a function of an image pickup apparatus.

In the zoom lens according to each example, the first lens unit (U1) may include a positive lens (U1Lp) at a position

closest to the image side. This makes it possible to realize a small zoom lens having a wide angle of view.

In the zoom lens according to each example, the lens unit (U2) having the strongest negative refractive power in the at least three movable lens units each of which moves during zooming may move to the image side during zooming from the wide-angle end to the telephoto end. This makes it possible to realize a small zoom lens having a high zoom ratio and high optical performance.

Hereinafter, a specific description is given of Examples 1 to 7. After the description of Example 7, Numerical Examples 1 to 7 corresponding to Examples 1 to 7 are given.

Example 1

A zoom lens according to Example 1 (Numerical Example 1) illustrated in FIG. 1 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a negative refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a positive refractive power, and a fifth lens unit U5 having a positive refractive power. FIG. 1 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 1 has a zoom ratio of 9.8 and satisfies the inequality (3).

The first lens unit U1 is a lens unit fixed (that does not move) during zooming. The second to fourth lens units U2 to U4 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fourth lens units U2 to U4 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 moves to the image side, and the third lens unit U3 and fourth lens unit U4 move to the object side and then move to the image side. The fourth lens unit U4 includes an aperture diaphragm SP at a position close to the object side. The fifth lens unit U5 is a final lens unit fixed during zooming. IP represents an image plane. An image plane of an image sensor (image pickup element or photoelectric conversion element) or a photosensitive surface of a silver halide film is disposed on the image plane IP.

In the first lens unit U1, a sub unit U11 closest to the object side and a sub unit U13 third closest to the object side are fixed sub units each of which is fixed during focusing, and a sub unit U12 second closest to the object side and a sub unit U14 closest to the image side are movable sub units each of which moves during focusing. As indicated by arrows below the movable sub units U12 and U14 in the drawing, during focusing from an infinite distance side to a close side, the movable sub unit U12 moves to the image side, and the movable sub unit U14 moves to the object side.

In Numerical Example 1, the positive lens U5Lp closest to the image side in the fifth lens unit U5 satisfies the inequalities (1) to (2), the inequality (4), and the inequality (6). The movable lens unit having the strongest negative refractive power in the movable lens units is the second lens unit U2. The second lens unit U2 satisfies the inequality (5).

The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

Example 2

A zoom lens according to Example 2 (Numerical Example 2) illustrated in FIG. 3 includes, in order from an object side to an image side, a first lens unit U1 having a positive

refractive power, a second lens unit U2 having a negative refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a positive refractive power, and a fifth lens unit U5 having a positive refractive power. FIG. 3 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 2 has a zoom ratio of 10.0 and satisfies the inequality (3). P in the drawing represents an optical element having no refractive power such as a prism.

The first lens unit U1 is a lens unit fixed during zooming. The second to fourth lens units U2 to U4 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fourth lens units U2 to U4 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 moves to the image side, and the third lens unit U3 and fourth lens unit U4 move to the object side and then move to the image side. The fourth lens unit U4 includes an aperture diaphragm SP at a position close to the object side. The fifth lens unit U5 is a final lens unit fixed during zooming.

In the first lens unit U1, a sub unit U1f closest to the object side is a fixed sub unit that is fixed during focusing, and a sub unit U12 second closest to the object side (closest to the image side) is a movable sub unit that moves during focusing. As indicated by an arrow below the movable sub unit U12 in the drawing, the movable sub unit U12 moves to the object side during focusing from an infinite distance side to a close side.

In Numerical Example 2, the positive lens U5Lp closest to the image side in the fifth lens unit U5 satisfies the inequalities (1) to (2), the inequality (4), and the inequality (6). The movable lens unit having the strongest negative refractive power in the movable lens units is the second lens unit U2. The second lens unit U2 satisfies the inequality (5).

The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

Example 3

A zoom lens according to Example 3 (Numerical Example 3) illustrated in FIG. 5 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a positive refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a negative refractive power, a fifth lens unit U5 having a positive refractive power, and a sixth lens unit U6 having a positive refractive power. FIG. 5 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 3 has a zoom ratio of 20.0 and satisfies the inequality (3).

The first lens unit U1 is a lens unit fixed during zooming. The second to fifth lens units U2 to U5 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fifth lens units U2 to U5 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 and the third lens unit U3 move to the image side, and the fourth lens unit U4 and the fifth lens unit U5 move to the object side and then move to the image side. The fifth lens unit U5 includes an aperture diaphragm SP at a position close to the image side. The sixth lens unit U6 is a final lens unit fixed during zooming.

In the first lens unit U1, a sub unit U1f closest to the object side is a fixed sub unit that is fixed during focusing, and a

sub unit U12 second closest to the object side (closest to the image side) is a movable sub unit that moves during focusing. As indicated by an arrow below the movable sub unit U12 in the drawing, the movable sub unit U12 moves to the object side during focusing from an infinite distance side to a close side.

In Numerical Example 3, a positive lens U6Lp closest to the image side in the sixth lens unit U6 satisfies the inequalities (1) to (2), the inequality (4), and the inequality (6). The movable lens unit having the strongest negative refractive power in the movable lens units is the third lens unit U3. The third lens unit U3 satisfies the inequality (5).

The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

Example 4

A zoom lens according to Example 4 (Numerical Example 4) illustrated in FIG. 7 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a negative refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a positive refractive power, and a fifth lens unit U5 having a positive refractive power. FIG. 7 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 4 has a zoom ratio of 17.0 and satisfies the inequality (3).

The first lens unit U1 is a lens unit fixed during zooming. The second to fourth lens units U2 to U4 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fourth lens units U2 to U4 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 moves to the image side, and the third lens unit U3 and fourth lens unit U4 move to the object side and then move to the image side. The fourth lens unit U4 includes an aperture diaphragm SP at a position close to the object side. The fifth lens unit U5 is a final lens unit fixed during zooming.

In the first lens unit U1, a sub unit U1f closest to the object side is a fixed sub unit that is fixed during focusing, and a sub unit U12 second closest to the object side (closest to the image side) is a movable sub unit that moves during focusing. As indicated by an arrow below the movable sub unit U12 in the drawing, the movable sub unit U12 moves to the object side during focusing from an infinite distance side to a close side.

In Numerical Example 4, the positive lens U5Lp closest to the image side in the fifth lens unit U5 satisfies the inequalities (1) to (2), the inequality (4), and the inequality (6). The movable lens unit having the strongest negative refractive power in the movable lens units is the second lens unit U2. The second lens unit U2 satisfies the inequality (5).

The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

Example 5

A zoom lens according to Example 5 (Numerical Example 5) illustrated in FIG. 9 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a negative refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a positive

refractive power, and a fifth lens unit U5 having a positive refractive power. FIG. 9 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 5 has a zoom ratio of 4.5 and satisfies the inequality (3).

The first lens unit U1 is a lens unit fixed during zooming. The second to fourth lens units U2 to U4 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fourth lens units U2 to U4 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 moves to the image side, and the third lens unit U3 and fourth lens unit U4 move to the object side and then move to the image side. The fourth lens unit U4 includes an aperture diaphragm SP at a position close to the object side. The fifth lens unit U5 is a final lens unit fixed during zooming.

In the first lens unit U1, a sub unit U1f closest to the object side and a sub unit U12 second closest to the object side (closest to the image side) are fixed sub units each of which is fixed during focusing, and a sub unit U12 second closest to the object side is a movable sub unit that moves during focusing. As indicated by an arrow below the movable sub unit U12 in the drawing, the movable sub unit U12 moves to the image side during focusing from an infinite distance side to a close side.

In Numerical Example 5, the positive lens U5Lp closest to the image side in the fifth lens unit U5 satisfies the inequalities (1) to (2), the inequality (4), and the inequality (6). The movable lens unit having the strongest negative refractive power in the movable lens units is the second lens unit U2. The second lens unit U2 satisfies the inequality (5).

The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

Example 6

A zoom lens according to Example 6 (Numerical Example 6) illustrated in FIG. 11 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a negative refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a negative refractive power, a fifth lens unit U5 having a positive refractive power, and a sixth lens unit U6 having a positive refractive power. FIG. 11 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 6 has a zoom ratio of 40.0 and satisfies the inequality (3).

The first lens unit U1 is a lens unit fixed during zooming. The second to fifth lens units U2 to U5 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fifth lens units U2 to U5 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 and the third lens unit U3 move to the image side and the fourth lens unit U4 and the fifth lens unit U5 move to the object side and then move to the image side. The sixth lens unit U6 is a final lens unit fixed during zooming and includes an aperture diaphragm SP at a position close to the object side.

In the first lens unit U1, a sub unit U1f closest to the object side is a fixed sub unit that is fixed during focusing, and a sub unit U12 second closest to the object side (closest to the image side) is a movable sub unit that moves during focusing. As indicated by an arrow below the movable sub unit

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U12 in the drawing, the movable sub unit U12 moves to the object side during focusing from an infinite distance side to a close side.

In Numerical Example 6, a positive lens U6Lp closest to the image side in the sixth lens unit U6 satisfies the inequalities (1) to (2), the inequality (4) and the inequality (6). The movable lens unit having the strongest negative refractive power in the movable lens units is the third lens unit U3. The third lens unit U3 satisfies the inequality (5).

The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

Example 7

A zoom lens according to Example 7 (Numerical Example 7) illustrated in FIG. 13 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a negative refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a positive refractive power, and a fifth lens unit U5 having a positive refractive power. FIG. 13 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 7 has a zoom ratio of 13.33 and satisfies the inequality (3).

The first lens unit U1 is a lens unit fixed during zooming. The second to fourth lens units U2 to U4 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fourth lens units U2 to U4 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 moves to the image side, and the third lens unit U3 and fourth lens unit U4 move to the object side and then move to the image side. The fifth lens unit U5 is a final lens unit fixed during zooming and includes an aperture diaphragm SP at a position close to the object side.

In the first lens unit U1, a sub unit U1f closest to the object side and a sub unit U13 third closest to the object side (closest to the image side) are fixed sub units each of which is fixed during focusing, and a sub unit U12 second closest to the object side is a movable sub unit that moves during focusing. As indicated by an arrow below the movable sub unit U12 in the drawing, the movable sub unit U12 moves to the image side during focusing from an infinite distance side to a close side.

In Numerical Example 7, the positive lens U5Lp closest to the image side in the fifth lens unit U5 satisfies the inequalities (1) to (2) and the inequality (6) but does not satisfy the inequality (4). The movable lens unit having the strongest negative refractive power in the movable lens units is the second lens unit U2. The second lens unit U2 satisfies the inequality (5).

The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

FIGS. 2A, 4A, 6A, 8A, 10A, 12A, and 14A illustrate longitudinal aberrations (spherical aberration, astigmatism, distortion, and chromatic aberration) of the respective zoom lenses according to Numerical Examples 1 to 7 each of which is at the wide-angle end and focuses on an object at an infinite distance. FIGS. 2B, 4B, 6B, 8B, 10B, 12B, and 14B illustrate longitudinal aberrations (spherical aberration, astigmatism, distortion, and chromatic aberration) of the respective zoom lenses according to Numerical Examples 1 to 7 each of which is at the telephoto end and focuses on an

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object at an infinite distance. In each spherical aberration diagram, Fno represents an F number. A solid line represents spherical aberration at a d-line (wavelength 587.6 nm), a two-dot chain line represents spherical aberration at a g-line (wavelength 435.8 nm), and a one-dot chain line represents spherical aberration at a C-line (wavelength 656.3 nm). In each astigmatism diagram, a solid line S represents a sagittal image plane, and a broken line M represents a meridional image plane. Each distortion diagram illustrates distortion at the d-line. In each chromatic aberration diagram, a two-dot chain line represents lateral chromatic aberration at the g-line, and a one-dot chain line represents lateral chromatic aberration at the C-line. ω represents a half angle of view ($^{\circ}$).

Hereinafter, Numerical Examples 1 to 7 are given. In each Numerical Example, a surface number i represents an order of a surface counted from the object side. r represents a curvature radius (mm) of an i-th surface counted from the object side, d represents a lens thickness or an air distance (mm) between the i-th and an (i+1)-th surface, and nd represents a refractive index at the d-line of an optical material between the i-th surface and the (i+1)-th surface. vd represents an Abbe number at the d-line of the optical material between the i-th surface and the (i+1)-th surface. BF represents a back focus (mm). A “back focus” is an air conversion length of a distance on an optical axis from a final surface (a lens surface closest to the image side) of the zoom lens to a paraxial image plane. A “total lens length” is a length acquired by adding the back focus to a distance on the optical axis from a front surface (a lens surface closest to the object side) of the zoom lens to the final surface.

The Abbe number vd is expressed by the following equation where Nd, NF, and NC respectively represent refractive indexes at the d-line (587.6 nm), an F-line (486.1 nm), and the C-line (656.3 nm) of Fraunhofer lines.

$$vd = (Nd - 1) / (NF - NC)$$

“*” attached to the surface number represents that the surface has an aspherical shape. The aspherical shape is expressed by the following equation where, when an optical axis direction is assumed to be an X axis, a direction orthogonal to the optical axis is assumed to be an H axis, and a light traveling direction is assumed to be positive, R represents a paraxial curvature radius, K represents a conical constant, A3, A4, A5, A6, A7, A8, A9, A10, A11, A12, A13, A14, A15, and A16 represent aspherical coefficients.

$$X = \frac{\frac{H^2}{R}}{1 + \sqrt{1 - (1 + k)(H/R)^2}} + A4H^4 + A6H^6 + A8H^8 + A10H^{10} + A12H^{12} + A14H^{14} + A16H^{16} + A3H^3 + A5H^5 + A7H^7 + A9H^9 + A11H^{11} + A13H^{13} + A15H^{15}$$

In conic constants and aspherical coefficients, “e-x” represents $\times 10^{-x}$.

Table 1 summarizes values of each inequality in Numerical Examples 1 to 7.

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Numerical Example 1

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Unit: mm				
SURFACE DATA				
Surface Number	r	d	nd	vd
1	-291.688	2.80	1.80100	35.0
2	144.078	1.21		
3	153.543	4.43	1.95906	17.5
4	243.217	1.68		
5	168.214	12.15	1.60311	60.6
6*	-244.858	10.85		
7	173.088	2.50	1.83400	37.2
8	65.624	10.67	1.43875	94.7
9	340.161	6.61		
10	93.619	10.70	1.43387	95.1
11	-422.601	0.20		
12	84.501	9.16	1.59522	67.7
13	-5841.261	(Variable)		
14	305.506	0.95	1.75500	52.3
15	18.046	7.33		
16	-32.262	0.75	1.49700	81.5
17	37.702	6.51	1.83400	37.2
18	-25.260	0.43		
19	-23.116	1.20	1.77250	49.6
20*	-314.961	(Variable)		
21	-50.584	4.31	1.80810	22.8
22	-22.449	1.10	1.80100	35.0
23	-242.799	(Variable)		
24	∞	2.00		
(Diaphragm)				
25*	60.959	7.61	1.64000	60.1
26	-74.522	0.19		
27	54.596	1.10	1.85478	24.8
28	34.089	4.63	1.48749	70.2
29	78.845	(Variable)		
30	227.612	2.57	1.48749	70.2
31	-207.293	1.20	2.00100	29.1
32	57.206	0.98		
33	53.608	5.46	1.83400	37.2
34	-137.066	43.57		
35	53.272	6.12	1.43875	94.7
36	-288.236	0.96		
37	40.968	11.95	1.80810	22.8
38	-36.607	0.90	1.95375	32.3
39	26.862	4.06		
40	48.363	9.15	1.54814	45.8
41	-28.327	1.00	2.00100	29.1
42	78.725	-0.10		
43	49.073	10.57	1.49700	81.5
44	-30.061	(Variable)		
Image Plane	∞			
ASPHERICAL DATA				
6th Surface				
K = -5.87800e+001	A4 = -4.11338e-007		A6 = 2.06301e-010	
A8 = -8.94756e-014	A10 = 2.51858e-017		A12 = -3.21128e-021	
20th Surface				
K = 4.38678e+002	A4 = -4.89448e-006		A6 = -3.86929e-009	
A8 = 7.15676e-011	A10 = -6.47860e-013		A12 = 2.11048e-015	
25th Surface				
K = -5.34393e-001	A4 = -2.54409e-006		A6 = 5.82081e-010	
A8 = 1.79765e-012	A10 = -6.91154e-015		A12 = 7.30550e-018	
VARIOUS DATA				
ZOOM RATIO	9.80 Wide-Angle	Middle	Telephoto	
Focal length:	25.50	78.17	250.00	
F-Number:	2.79	2.80	3.70	
Half Angle of View (°):	30.13	10.72	3.39	
Image Height:	14.80	14.80	14.80	
Total Lens Length:	316.95	316.95	316.95	

-continued

Unit: mm			
BF:	37.94	37.94	37.94
d13	0.99	38.85	59.24
d20	48.23	3.58	2.77
d23	9.62	18.81	0.97
d29	10.71	8.31	6.56
d44	37.94	37.94	37.94
LENS UNIT DATA			
Unit	Starting Surface		Focal length
1	1		90.20
2	14		-22.61
3	21		-82.03
4	24		54.87
5	30		102.85

Numerical Example 2

Unit: mm				
SURFACE DATA				
Surface Number	r	d	nd	vd
1	-175.214	3.50	1.67300	38.3
2	151.202	15.58		
3	-119.783	3.61	1.43387	95.1
4	-112.613	0.20		
5	227.415	13.96	1.43387	95.1
6	-129.184	9.38		
7	162.176	8.02	1.43387	95.1
8	-753.576	0.08		
9	125.294	8.39	1.43387	95.1
10	-915.044	0.83		
11	74.473	8.42	1.43875	94.7
12*	340.725	(Variable)		
13	269.294	1.30	2.00069	25.5
14	20.440	6.36		
15	-39.720	1.10	1.69680	55.5
16	32.336	6.99	1.85478	24.8
17	-29.629	1.02		
18	-23.763	1.10	1.72916	54.7
19*	-81.231	(Variable)		
20	-48.638	1.30	1.88300	40.8
21	208.585	2.06	1.95906	17.5
22	-213.735	(Variable)		
23*	73.373	3.79	1.72916	54.7
24	-49.264	(Variable)		
25	∞	6.28		
(Diaphragm)				
26	-44.725	3.83	1.65412	39.7
27	-19.881	1.20	1.61800	63.3
28	43.010	2.37		
29	33.497	1.30	1.96300	24.1
30	22.812	6.38	1.65412	39.7
31	-58.802	15.84		
32	71.434	3.35	1.95906	17.5
33	103.023	1.30	2.00100	29.1
34	28.089	1.40		
35	31.130	9.86	1.43875	94.7
36	-22.965	1.40	2.00100	29.1
37	-74.760	4.89	1.49700	81.5
38	-38.132	5.45		
39	-83.722	6.77	1.59270	35.3
40	-34.630	5.12		
41	128.652	5.93	1.49700	81.5
42	-100.375	5.00		

-continued

Unit: mm				
43	∞	63.04	1.60859	46.4
44	∞	8.70	1.51633	64.2
45	∞	(Variable)		
Image Plane	∞			
ASPHERICAL DATA				
12th Surface				
K = 0.00000e+000	A4 = 8.27789e-008	A6 = -1.49973e-011		
A8 = 4.85422e-015	A10 = -5.95615e-019			
19th Surface				
K = 1.35528e+000	A4 = -6.45504e-006	A6 = -5.00746e-009		
A8 = -1.21703e-010	A10 = 4.12728e-012	A12 = -6.25642e-014		
A14 = 4.60921e-016	A16 = 1.33684e-018			
23rd Surface				
K = 1.92930e+000	A4 = -4.90734e-006	A6 = 5.95339e-009		
A8 = -1.41239e-010	A10 = 2.19152e-012	A12 = -1.73996e-014		
A14 = 6.52571e-017	A16 = -8.36889e-020			
VARIOUS DATA				
ZOOM RATIO	10.00 Wide-Angle	Middle	Telephoto	
Focal length:	14.00	43.48	140.00	
F-Number:	2.90	2.90	2.90	
Half Angle of View (°):	33.45	12.01	3.78	
Image Height:	9.25	9.25	9.25	
Total Lens Length:	334.12	334.12	334.12	
BF:	16.63	16.63	16.63	
d12	1.50	38.69	58.72	
d19	57.19	10.12	0.63	
d22	0.99	10.68	0.97	
d24	1.39	1.58	0.74	
d45	16.63	16.63	16.63	
LENS UNIT DATA				
Unit	Starting Surface	Focal length		
1	1	77.73		
2	13	-19.98		
3	20	-75.96		
4	23	40.78		
5	25	59.39		

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Numerical Example 3

-continued

Unit: mm					Unit: mm				
SURFACE DATA									
Surface Number	r	d	nd	vd					
1	-171.783	2.30	1.80100	35.0	50	16	-64.571	7.14	1.80810
2	360.342	3.32				17	-13.488	0.70	1.88300
3	2449.824	2.30	1.80100	35.0		18	76.051	0.20	
4	93.547	17.69	1.49700	81.5		19	26.817	2.89	1.66680
5	-127.730	0.40				20	63.291	(Variable)	
6	164.755	8.90	1.43387	95.1	55	21	-43.870	0.75	1.77250
7	-323.842	6.90				22	30.844	3.77	1.69895
8	119.088	10.53	1.61800	63.3		23	-80.916	2.22	
9	-260.350	0.15				24	-28.353	0.75	1.77250
10	65.966	5.53	1.77250	49.6		25	-60.179	(Variable)	
11	113.238	(Variable)				26	-122.743	3.62	1.63854
12	108.681	2.33	1.77250	49.6	60	27	-33.339	0.15	
13	149.799	(Variable)				28	-178.655	2.66	1.51633
14	67.372	0.90	1.88300	40.8	65	29	-58.384	(Variable)	
15	14.158	6.41				30	∞	1.30	
						(Diaphragm)			
						31	84.875	6.24	1.51742
						32	-30.166	0.90	1.83481
						33	-88.784	32.40	
						34	82.335	5.14	1.49700
						35	-52.958	9.14	81.5

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-continued

Unit: mm				
36	-451.214	1.40	1.83403	37.2
37	24.017	5.09	1.48749	70.2
38	137.787	0.36		
39	64.625	7.10	1.50127	56.5
40	-22.356	1.40	1.83481	42.7
41	-69.828	3.34		
42	71.892	5.20	1.48749	70.2
43	-44.146	0.50		
44	-86.855	2.10	1.49700	81.5
45	-64.323	4.00		
46	∞	33.00	1.60859	46.4
47	∞	13.20	1.51633	64.1
48	∞	(Variable)		
Image Plane	∞			

VARIOUS DATA

ZOOM RATIO	20.00 Wide-Angle	Middle	Telephoto
Focal length:	8.00	35.00	160.00
F-Number:	1.90	1.90	2.80
Half Angle of View (°):	34.51	8.76	1.97
Image Height:	5.50	5.50	5.50
Total Lens Length:	295.50	295.50	295.50
BF:	7.50	7.50	7.50
d11	0.83	26.24	37.12
d13	0.80	12.02	16.83
d20	54.59	8.56	7.32
d25	6.51	11.46	1.47
d29	0.99	5.45	0.98
d48	7.50	7.50	7.50

LENS UNIT DATA

Unit	Starting Surface	Focal length
1	1	71.25
2	12	500.20
3	14	-14.23
4	21	-38.15
5	26	49.99
6	30	57.02

Numerical Example 4

Unit: mm				
SURFACE DATA				
Surface Number	r	d	nd	vd
1	-161.563	2.20	1.74951	35.3
2	83.531	1.54		
3	89.927	6.77	1.43387	95.1
4	183.771	0.20		
5	135.868	15.24	1.43387	95.1
6	-95.350	7.88		
7	136.897	6.93	1.43387	95.1
8	-454.092	0.15		
9	91.750	8.76	1.43387	95.1

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-continued

Unit: mm				
10	-283.987	0.15		
11	53.445	4.48	1.76385	48.5
12	88.487	(Variable)		
13	70.773	0.90	1.88300	40.8
14	13.760	5.02		
15	-66.538	6.70	1.80810	22.8
16	-12.185	0.70	1.88300	40.8
17	52.502	0.20		
18	24.812	2.78	1.62588	35.7
19	88.435	(Variable)		
20	-26.441	0.70	1.83400	37.2
21	27.074	3.42	1.84649	23.9
22	-190.460	(Variable)		
23	∞	1.96		

(Diaphragm)

24	-83.419	3.00	1.63854	55.4
25	-32.030	0.15		
26	11679.132	2.19	1.51823	58.9
27	-115.244	0.07		
28	68.450	5.93	1.51742	52.4
29	-31.378	0.90	1.74951	35.3
30	-76.848	(Variable)		
31	274.509	2.76	1.49700	81.5
32	-96.339	6.26		
33	60.490	1.40	1.80100	35.0
34	22.542	5.43	1.48749	70.2
35	76.507	0.11		
36	59.172	7.71	1.50127	56.5
37	-23.847	1.40	1.83481	42.7
38	-96.461	0.15		
39	100.150	5.98	1.43875	94.7
40	-32.303	4.00		
41	∞	33.00	1.60859	46.4
42	∞	13.20	1.51633	64.1
43	∞	(Variable)		
Image Plane	∞			

VARIOUS DATA

ZOOM RATIO	17.00 Wide-Angle	Middle	Telephoto
Focal length:	8.00	32.65	136.00
F-Number:	1.90	1.90	2.50
Half Angle of View (°):	34.51	9.56	2.32
Image Height:	5.50	5.50	5.50
Total Lens Length:	271.31	271.31	271.31
BF:	7.64	7.64	7.64
d12	0.79	32.80	47.01
d19	45.91	9.49	12.81
d22	6.27	10.96	1.15
d30	40.35	40.09	32.36
d43	7.64	7.64	7.64

LENS UNIT DATA

Unit	Starting Surface	Focal length
1	1	61.00
2	13	-13.50
3	20	-37.99
4	23	37.71
5	31	52.42

Numerical Example 5

Unit: mm				
SURFACE DATA				
Surface Number	r	d	nd	vd
1*	78.953	2.80	1.77250	49.6
2	42.025	21.29		

-continued

Unit: mm				
3	68.913	2.20	1.77250	49.6
4	44.989	20.10		
5	353.478	2.20	1.77250	49.6
6	102.861	4.67	1.92286	18.9
7	143.293	10.00		
8	143.296	8.54	1.49700	81.5
9*	-660.058	10.16		
10	126.072	12.33	1.61800	63.3
11	-89.566	2.02		
12	-385.544	2.00	1.80518	25.4
13	67.762	3.01	1.49700	81.5
14	66.178	4.38		
15	93.903	10.00	1.59522	67.7
16	-116.798	0.20		
17	79.466	7.23	1.73000	49.0
18	-204.448	(Variable)		
19*	1243.240	1.30	1.77250	49.6
20	23.783	10.00		
21	-42.047	0.90	1.77250	49.6
22	29.864	4.04	1.84666	23.8
23	-71.594	2.91		
24	-18.875	0.90	1.80100	35.0
25	-26.589	(Variable)		
26	-33.939	0.90	1.72916	54.7
27	54.755	2.93	1.84666	23.8
28	1294.668	(Variable)		
29	∞	1.85		
(Diaphragm)				
30	700.247	4.17	1.81600	46.6
31	-56.385	0.20		
32	320.448	3.94	1.51742	52.4
33	-53.596	(Variable)		
34	53.543	8.45	1.48749	70.2
35	-35.854	1.20	2.00069	25.5
36	-224.787	0.20		
37	28.978	3.77	1.53172	48.8
38	35.232	20.70		
39	265.405	6.25	1.51823	58.9
40	-64.862	14.34		
41	109.038	8.83	1.49700	81.5
42	-21.348	0.85	1.95375	32.3
43	-120.445	2.16		
44	1741.197	6.19	1.92286	18.9
45	-34.707	0.85	2.00100	29.1
46	100.346	1.04		
47	49.126	6.84	1.49700	81.5
48	-56.043	(Variable)		
Image Plane	∞			
ASPHERICAL DATA				
1st Surface				
K = 6.16535e-001 A8 = -3.07226e-015	A4 = 9.53347e-008	A6 = 2.00290e-011		
9th Surface				
K = -8.09015e+002 A8 = -1.44173e-013	A4 = 7.34432e-007	A6 = 1.86751e-010		
19th Surface				
K = -1.93517e+004 A8 = 2.17579e-011	A4 = 8.47692e-006	A6 = -7.51813e-009		
VARIOUS DATA				
4.50				
ZOOM RATIO	Wide-Angle	Middle	Telephoto	
Focal length:	16.00	33.31	71.93	
F-Number:	2.99	3.00	3.00	
Half Angle of View (°):	44.18	25.03	12.20	
Image Height:	15.55	15.55	15.55	
Total Lens Length:	320.01	320.01	320.01	
BF:	39.96	39.96	39.96	
d18	1.00	17.78	30.19	
d25	29.70	7.90	2.00	
d28	2.74	5.44	1.24	

-continued

Unit: mm			
d33	7.77	10.08	7.77
d48	39.96	39.96	39.96
LENS UNIT DATA			
Unit	Starting Surface	Focal length	
1	1	32.00	
2	19	-18.80	
3	26	-50.00	
4	29	37.74	
5	34	95.79	

Numerical Example 6

Unit: mm				
SURFACE DATA				
Surface Number	r	d	nd	vd
1	234.095	7.49	1.43387	95.1
2	744.347	1.00		
3	611.406	3.00	1.80100	35.0
4	179.646	1.07		
5	180.854	14.73	1.43387	95.1
6	-616.575	11.15		
7	206.490	7.85	1.43387	95.1
8	1434.650	0.20		
9	143.338	5.46	1.43387	95.1
10	229.576	0.20		
11	118.792	8.23	1.43387	95.1
12	293.076	(Variable)		
13	157.514	1.00	1.83400	37.2
14	27.285	8.17		
15	-53.185	0.90	1.77250	49.6
16	48.123	0.70		
17	39.025	7.61	1.80810	22.8
18	-58.658	(Variable)		
19	-36.219	1.10	1.77250	49.6
20	-16640.477	(Variable)		
21	-62.809	1.30	1.71300	53.9
22	37.542	3.41	1.83400	37.2
23	123.493	(Variable)		
24	622.004	3.95	1.74951	35.3
25	-65.214	1.00		
26	-78.813	1.53	1.51823	58.9
27	-74.268	(Variable)		
28	∞	1.00		
(Diaphragm)				
29	91.777	9.43	1.74951	35.3
30	-37.804	1.50	2.00100	29.1
31	174.214	1.00		
32	84.520	8.31	1.48749	70.2
33	-39.356	1.50	1.88300	40.8
34	-60.994	50.00		
35	163.915	4.64	1.51742	52.4
36	-55.320	2.54		
37	36.082	1.20	1.78590	44.2

-continued

Unit: mm					
38	15.322	6.50	1.51742	52.4	
39	69.483	2.01			
40	100.129	5.44	1.54814	45.8	
41	-2364.899	1.20	1.77250	49.6	
42	29.077	0.67			
43	20.807	4.04	1.49700	81.5	
44	614.656	3.80			
45	∞	34.37	1.60859	46.4	
46	∞	13.75	1.51680	64.2	
47	∞	(Variable)			
Image Plane	∞				
VARIOUS DATA					
	40.00				
ZOOM RATIO	Wide-Angle	Middle	Telephoto		
Focal length:	11.00	218.22	440.00		
F-Number:	2.10	2.10	4.10		
Half Angle of View (°):	26.57	1.44	0.72		
Image Height:	5.50	5.50	5.50		
Total Lens Length:	402.14	402.14	402.14		
BF:	7.90	7.90	7.90		
d12	10.00	108.86	113.65		
d18	1.43	6.19	11.14		
d20	125.30	2.96	22.17		
d23	10.04	17.80	2.30		
d27	3.50	14.46	1.00		
d47	7.90	7.90	7.90		
LENS UNIT DATA					
Unit	Starting Surface		Focal length		
1	1		161.84		
2	13		-60.64		
3	19		-46.99		
4	21		-66.60		
5	24		76.81		
6	28		70.45		

Numerical Example 7

Unit: mm				
SURFACE DATA				
Surface Number	r	d	nd	vd
1*	1192.402	2.50	1.83481	42.7
2	30.426	16.96		
3*	137.578	2.00	1.83481	42.7
4	78.229	10.11		

-continued

Unit: mm				
5	-100.411	1.80	1.83481	42.7
6	-481.398	0.15		
7	93.872	3.85	1.92286	18.9
8	257.624	2.21		
9	179.156	7.63	1.60300	65.4
10*	-106.058	4.41		
11	4935.437	8.72	1.43387	95.1
12	-55.764	0.65		
13	-57.428	1.70	1.80000	29.8
14	-104.843	0.18		
15	173.505	1.70	1.91650	31.6
16	56.124	12.67	1.43875	94.7
17	-148.337	0.40		
18	4271.888	9.14	1.43387	95.1
19	-62.811	0.40		
20	143.432	7.53	1.76385	48.5
21	-146.520	(Variable)		
22	106.192	0.70	2.00100	29.1
23	18.267	4.25		
24	-54.292	0.70	1.43875	94.7
25	89.426	2.07		
26	-193.060	5.57	1.85478	24.8
27	-15.511	0.70	1.88300	40.8
28	140.345	0.21		
29	39.231	2.87	1.64769	33.8
30	-209.632	(Variable)		
31	-34.647	0.80	1.72916	54.7
32	49.836	2.41	1.84666	23.8
33	4117.439	(Variable)		
34*	59.371	6.11	1.58913	61.1
35	-59.288	(Variable)		
36	∞	2.45		
(Diaphragm)				
37	196.135	4.72	1.51742	52.4
38	-47.588	1.00	1.83481	42.7
39	-156.750	35.50		
40	62.539	5.60	1.63980	34.5
41	-48.447	1.79		
42	-70.201	0.90	1.88300	40.8
43	29.203	5.31	1.48749	70.2
44	-106.840	0.20		
45	64.763	7.66	1.43875	94.7
46	-21.285	0.90	2.00100	29.1
47	-59.870	0.13		
48	175.318	4.80	1.61800	63.3
49	-35.144	4.00		
50	∞	33.00	1.60859	46.4
51	∞	13.20	1.51680	64.2
52	∞	(Variable)		
Image Plane	∞			
ASPHERICAL DATA				
1st Surface				
K = 0.00000e+000	A4 = 4.79471e-006	A6 = 1.09101e-008		
A8 = 7.85275e-012	A10 = 9.49250e-014	A12 = 1.11174e-016		
A14 = 1.85317e-020	A16 = -6.47635e-026	A3 = 6.81313e-006		
A5 = -1.91840e-007	A7 = -3.04740e-010	A9 = -1.17561e-012		
A11 = -4.11498e-015	A13 = -1.90019e-018	A15 = -7.32675e-023		
3rd Surface				
K = 0.00000e+000	A4 = -2.88353e-006	A6 = -7.32735e-008		
A8 = -7.11623e-010	A10 = -3.23231e-013	A12 = 1.59758e-015		
A14 = -6.51939e-019	A16 = -2.03836e-022	A3 = -5.37483e-006		
A5 = 4.33123e-007	A7 = 9.09152e-009	A9 = 3.03607e-011		
A11 = -3.27297e-014	A13 = -1.78387e-017	A15 = 2.23171e-020		
10th Surface				
K = 0.00000e+000	A4 = 1.24811e-006	A6 = 1.37302e-008		
A8 = 2.71930e-010	A10 = 2.08643e-013	A12 = -7.68000e-016		
A14 = 1.05314e-018	A16 = 2.19108e-022	A3 = -1.60560e-006		
A5 = -1.67398e-008	A7 = -2.65872e-009	A9 = -1.44572e-011		
A11 = 1.68256e-014	A13 = -4.85868e-018	A15 = -2.64140e-020		

-continued

Unit: mm			
34th Surface			
K = -8.55475e+000	A4 = 1.00844e-006	A6 = -2.95364e-009	
A8 = 1.16661e-012			
VARIOUS DATA			
ZOOM RATIO	13.33 Wide-Angle	Middle	Telephoto
Focal length:	4.50	16.51	59.99
F-Number:	1.90	1.90	2.80
Half Angle of View (°):	50.71	18.43	5.24
Image Height:	5.50	5.50	5.50
Total Lens Length:	315.65	315.65	315.65
BF:	7.45	7.45	7.45
d21	0.65	37.62	53.47
d30	42.42	5.41	4.56
d33	15.45	18.86	2.11
d35	7.42	4.05	5.80
d52	7.45	7.45	7.45
LENS UNIT DATA			
Unit	Starting Surface	Focal length	
1	1	30.38	
2	22	-20.82	
3	31	-53.01	
4	34	51.14	
5	36	52.07	

TABLE 1

Ineq.	No.	Inequality	Example						
			1	2	3	4	5	6	7
(1) (4)	vdp		81.54	81.54	81.54	94.66	81.54	81.54	63.33
(2)	Ndp + 0.01vdp		2.31	2.31	2.31	2.39	2.31	2.31	2.25
(3)	Z		9.80	10.00	20.00	17.00	4.50	40.00	13.33
(5)	fp/fv		1.74	5.71	34.01	4.18	2.86	0.92	2.29
(6)	Le/fw		3.86	5.46	10.04	3.90	5.10	9.09	15.22

Image Pickup Apparatus

FIG. 15 illustrates a configuration of a camera system (image pickup apparatus) 125 that uses, as an image pickup optical system, the zoom lens according to any one of Examples 1 to 7. A reference numeral 101 denotes the zoom lens according to any one of Examples 1 to 7. A reference numeral 124 denotes a camera, and the zoom lens 101 is an interchangeable lens that is attachable to and detachable from the camera 124.

The zoom lens 101 includes a first lens unit F, a zooming unit LZ that moves during zooming, and imaging lens units R1 to R3. The first lens unit F includes a movable sub unit that moves during focusing. The zooming unit LZ includes at least three lens units each of which moves during zooming. An aperture diaphragm SP and the imaging lens units R1 to R3 are disposed on the image side of the zooming unit LZ. A lens unit IE, which is an extender or the like, is inserted between any two lens units of the imaging lens units R1 to R3 (for example, between lens units R1 and R2) from the outside of the optical path.

Reference numerals 114 and 115 denote driving mechanisms including, for example, a helicoid, a cam, and the like, that respectively drive the movable sub unit in the first lens unit F and the movable lens units in the zooming unit LZ in

the optical axis direction. Reference numerals 116, 117, and 118 are actuators that electrically drive the driving mechanisms 114 and 115 and the aperture diaphragm SP.

Reference numerals 119, 120, and 121 denote detectors such as encoder, potentiometers, and photosensors, for detecting positions on the optical axis of the movable sub unit in the first lens unit F and the movable lens units in the zooming unit LZ and an aperture diameter of the aperture diaphragm SP.

In the camera 124, a reference numeral 109 denotes a glass block corresponding to an optical filter or a color separation optical system, and a reference numeral 110 denotes an image sensor (image pickup element or photoelectric conversion element), such as a CCD sensor and a CMOS sensor, that captures (configured to pick up) an optical image (object image) of an object formed by the zoom lens 101.

Reference numerals 111 and 122 denote CPUs as controllers that control the driving of the camera 124 and the zoom lens 101.

When the zoom lens according to any one of Examples 1 to 7 is used as the image pickup optical system of the camera system 125, it is possible to realize high imaging performance from an optical aspect.

According to the above examples, it is possible to provide, for example, a zoom lens that is beneficial to size reduction, a wide angle of view, a high zoom ratio, a large diameter ratio, and high optical performance.

While the disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of Japanese Patent Application No. 2021-131582, filed on Aug. 12, 2021, which is hereby incorporated by reference herein in its entirety.

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What is claimed is:

1. A zoom lens comprising, in order from an object side to an image side:

a first lens unit having a positive refractive power and configured not to move for zooming;

at least three movable lens units configured to move so that a distance between each pair of adjacent lens units therein changes for zooming; and

a final lens unit disposed closest to the image side in lens units included in the zoom lens, having a positive refractive power, and configured not to move for zooming,

wherein a movable lens unit closest to the image side in the at least three movable lens units has a positive refractive power, and a movable lens unit second closest to the image side in the at least three movable lens units has a negative refractive power,

wherein the first lens unit includes:

a first sub unit closest to the object side and configured not to move for focusing; and

a second sub unit configured to move for focusing,

wherein the final lens unit includes a positive lens closest to the image side in elements having a refractive power in the final lens unit, and

wherein following inequalities are satisfied:

$$80 \leq vdp \leq 81.54,$$

$$2.24 \leq Ndp + 0.01 vdp,$$

$$4 \leq Z \leq 50, \text{ and}$$

$$2.5 \leq Le/fw \leq 5.46$$

where vdp represents an Abbe number with respect to a d-line of the positive lens, Ndp represents a refractive index with respect to the d-line of the positive lens, Z represents a zoom ratio of the zoom lens, Le represents a distance on an optical axis from a lens surface closest to the object side in the final lens unit to an image side surface of the positive lens in the final lens unit, and fw represents a focal length of the zoom lens at a wide-angle end.

2. The zoom lens according to claim 1, wherein the final lens unit includes four or more lenses each of which has a refractive power.

3. The zoom lens according to claim 1, wherein a following inequality is satisfied:

$$0.5 \leq |fp/fv| \leq 50.0$$

where fp represents a focal length of the positive lens, and fv represents a focal length of a movable lens unit having a strongest negative refractive power in the at least three movable lens units.

4. The zoom lens according to claim 1, wherein a movable lens unit having a strongest negative refractive power in the at least three movable lens units includes two negative lenses and one positive lens.

5. The zoom lens according to claim 1, wherein the first lens unit includes a positive lens at a position closest to the image side in the first lens unit.

6. The zoom lens according to claim 1, wherein a movable lens unit having a strongest negative refractive power in the at least three movable lens units moves to the image side for zooming from a wide-angle end to a telephoto end.

7. The zoom lens according to claim 1, wherein the zoom lens includes, as the at least three movable lens units, in order from the object side to the image side:

a second lens unit having a negative refractive power;

a third lens unit having a negative refractive power; and

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a fourth lens unit having a positive refractive power, and wherein the zoom lens includes, as the final lens unit, a fifth lens unit having a positive refractive power.

8. The zoom lens according to claim 1, wherein the zoom lens includes, as the at least three movable lens units, in order from the object side to the image side:

a second lens unit having a positive or negative refractive power;

a third lens unit having a negative refractive power;

a fourth lens unit having a negative refractive power; and

a fifth lens unit having a positive refractive power, wherein the zoom lens includes, as the final lens unit, a sixth lens unit having a positive refractive power.

9. An apparatus comprising:

a zoom lens; and

a pickup element configured to pick up an image formed by the zoom lens,

wherein the zoom lens includes, in order from an object side to an image side:

a first lens unit having a positive refractive power and configured not to move for zooming;

at least three movable lens units configured to move so that a distance between each pair of adjacent lens units therein changes for zooming; and

a final lens unit disposed closest to the image side in lens units included in the zoom lens, having a positive refractive power, and configured not to move for zooming,

wherein a movable lens unit closest to the image side in the at least three movable lens units has a positive refractive power, and a movable lens unit second closest to the image side in the at least three movable lens units has a negative refractive power,

wherein the first lens unit includes:

a first sub unit closest to the object side and configured not to move for focusing; and

a second sub unit configured to move for focusing,

wherein the final lens unit includes a positive lens closest to the image side in elements having a refractive power in the final lens unit,

wherein following inequalities are satisfied:

$$80 \leq vdp \leq 81.54,$$

$$2.24 \leq Ndp + 0.01 vdp,$$

$$4 \leq Z \leq 50, \text{ and}$$

$$2.5 \leq Le/fw \leq 5.46$$

where vdp represents an Abbe number with respect to a d-line of the positive lens, Ndp represents a refractive index with respect to the d-line of the positive lens, Z represents a zoom ratio of the zoom lens, Le represents a distance on an optical axis from a lens surface closest to the object side in the final lens unit to an image side surface of the positive lens in the final lens unit, and fw represents a focal length of the zoom lens at a wide-angle end.

10. The apparatus according to claim 9, wherein the final lens unit includes four or more lenses each of which has a refractive power.

11. The apparatus according to claim 9, wherein a following inequality is satisfied:

$$0.5 \leq |fp/fv| \leq 50.0$$

where fp represents a focal length of the positive lens, and fv represents a focal length of a movable lens unit having a strongest negative refractive power in the at least three movable lens units.

12. The apparatus according to claim 9, wherein a movable lens unit having a strongest negative refractive power in the at least three movable lens units includes two negative lenses and one positive lens.

13. The apparatus according to claim 9, wherein the first lens unit includes a positive lens at a position closest to the image side in the first lens unit.

14. The apparatus according to claim 9, wherein a movable lens unit having a strongest negative refractive power in the at least three movable lens units moves to the image side for zooming from a wide-angle end to a telephoto end.

15. The apparatus according to claim 9, wherein the zoom lens includes, as the at least three movable lens units, in order from the object side to the image side:

- a second lens unit having a negative refractive power;
- a third lens unit having a negative refractive power; and
- a fourth lens unit having a positive refractive power, and wherein the zoom lens includes, as the final lens unit, a fifth lens unit having a positive refractive power.

16. The apparatus according to claim 9, wherein the zoom lens includes, as the at least three movable lens units, in order from the object side to the image side:

- a second lens unit having a positive or negative refractive power;
- a third lens unit having a negative refractive power;
- a fourth lens unit having a negative refractive power; and
- a fifth lens unit having a positive refractive power, wherein the zoom lens includes, as the final lens unit, a sixth lens unit having a positive refractive power.

17. A zoom lens comprising, in order from an object side to an image side:

- a first lens unit having a positive refractive power and configured not to move for zooming;
- at least three movable lens units configured to move so that a distance between each pair of adjacent lens units therein changes for zooming; and

a final lens unit disposed closest to the image side in lens units included in the zoom lens, having a positive refractive power, and configured not to move for zooming,

wherein a movable lens unit closest to the image side in the at least three movable lens units has a positive refractive power, and a movable lens unit second closest to the image side in the at least three movable lens units has a negative refractive power,

wherein a movable lens unit having a strongest negative refractive power in the at least three movable lens units includes two negative lenses and one positive lens,

wherein the first lens unit includes:

- a first sub unit closest to the object side and configured not to move for focusing; and
- a second sub unit configured to move for focusing,

wherein the final lens unit includes a positive lens closest to the image side in elements having a refractive power in the final lens unit, and

wherein following inequalities are satisfied:

$$62 \leq vdp \leq 81.54,$$

$$2.24 \leq Ndp + 0.01 vdp,$$

$$4 \leq Z \leq 50, \text{ and}$$

$$2.5 \leq Le/fw \leq 5.46$$

where vdp represents an Abbe number with respect to a d-line of the positive lens, Ndp represents a refractive index with respect to the d-line of the positive lens, Z represents a zoom ratio of the zoom lens, Le represents a distance on an optical axis from a lens surface closest to the object side in the final lens unit to an image side surface of the positive lens in the final lens unit, and fw represents a focal length of the zoom lens at a wide-angle end.

18. An apparatus comprising:

- a zoom lens; and
 - a pickup element configured to pick up an image formed by the zoom lens,
- wherein the zoom lens includes, in order from an object side to an image side:

- a first lens unit having a positive refractive power and configured not to move for zooming;
- at least three movable lens units configured to move so that a distance between each pair of adjacent lens units therein changes for zooming; and
- a final lens unit disposed closest to the image side in lens units included in the zoom lens, having a positive refractive power, and configured not to move for zooming,

wherein a movable lens unit closest to the image side in the at least three movable lens units has a positive refractive power, and a movable lens unit second closest to the image side in the at least three movable lens units has a negative refractive power,

wherein a movable lens unit having a strongest negative refractive power in the at least three movable lens units includes two negative lenses and one positive lens,

wherein the first lens unit includes:

- a first sub unit closest to the object side and configured not to move for focusing; and
- a second sub unit configured to move for focusing,

wherein the final lens unit includes a positive lens closest to the image side in elements having a refractive power in the final lens unit,

wherein following inequalities are satisfied:

$$62 \leq vdp \leq 81.54,$$

$$2.24 \leq Ndp + 0.01 vdp,$$

$$4 \leq Z \leq 50, \text{ and}$$

$2.5 \leq Le/fw \leq 5.46$ ere vdp represents an Abbe number with respect to a d-line of the positive lens, Ndp represents a refractive index with respect to the d-line of the positive lens, Z represents a zoom ratio of the zoom lens, Le represents a distance on an optical axis from a lens surface closest to the object side in the final lens unit to an image side surface of the positive lens in the final lens unit, and fw represents a focal length of the zoom lens at a wide-angle end.

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